

Basic Ecological Concepts and Data in R

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2026-01-14

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Preface

This book provides an activity-based approach introduction to the science and practice of ecology. It is designed to form the basis of an undergraduate general ecology course. It includes real-world assessment activities, and so answer keys are not generally available, even if they might be appropriate in some situations.

Each chapter provides,

- links to background information in the form of a video lecture, peer-reviewed publications, and sometimes other sources,
- one or more writing assignments, and
- data exploration and modeling.

Each chapter mixes (i) ecological content and (ii) coding practices. Students will get the most out of the content if they understand and *accept* this from the start. Avoiding one or the other reduces their ability to understand, appreciate, and use both.

Guidance

Use these tips to help reduce frustration and make things work better:

- The book should be read and the code implemented in sequence. For example, you should complete chapter 1 before chapter 2, complete chapter 2 before chapter 3, and so on. That is for two reasons. First, you will learn things about R and your computer in earlier chapters that you may need to know for later chapters. This is especially true for the first few chapters. Second, you will gain ecological knowledge in earlier chapters that enrich your experience in later chapters.
- When the text provides an enumerated list or *numbered* bullet items (e.g., 1, 2, 3 or a, b, c), it is usually very important or even essential to do these steps in sequence. There will be times that it is impossible to succeed unless you do the steps in *sequence*.
- When the text provides a simple, unnumbered list, it is usually not important to consider these items as representing a sequence.
- I included a few sections entitled *Overthinking*. I stole this idea from Richard McElreath (McElreath 2020), and these are sections that you can skip on the first read, and are not essentially for getting things to *work*. They provide deeper, more detailed explanations than is typically required.

Other books

My other books include *[A Primer of Ecology using R](#)* and *[A Primer of Ecosystem Modeling](#)*. The former (Primer of Ecology) is designed primarily for beginning graduate students as a primer to theory in population and community ecology. It was first published with Springer in 2009, and the current GitHub version is quite updated. I use the latter (ecosystem modeling) when I co-instruct a graduate course led by colleagues who are real ecosystem ecologists (Drs. Melany Fisk, Mike Vanni, and Lesley Knoll).

Mazel tov!

1 Computer setup and first R script

1.1 Goals

- Learn about your computer and how to organize its files.
- Install R, an open source programming language.
- Begin to use R effectively.

1.2 Background

In this project, you will begin to use tools for data exploration and analysis. This tools and your computer are really important for helping you make sense of our really complicated world. Ecology is way more complicated than physics or engineering, and the data are noisy and often unpredictable. We need to use computers to make sense of it.

We will use open source and free software to help us understand ecological systems. You will download and install R, known formally as the R programming language and environment. You will also install RStudio, an integrated development environment (IDE), designed to be used with R. The R interface looks a little different on Windows, Macs, and Linux computers. RStudio will allow us all share an identical appearance and user experience. You will use R by opening RStudio.

In this class, you will use the English language to write about ecology, and the R language to explore and analyze ecological data.

1.2.1 Understanding you computer

First, we need to make sure we know some basics about computers.

1.2.2 Folders, directories, and directory structure

On your computer,

- a folder is a place to put files and other folders,

- “a directory is a file system cataloging structure that contains references to other computer files, and possibly other directories.” (Wikipedia)
- A folder is a part of your computer’s directory system.

If you have not already done so, **you need to organize your computer.**

Do these:

- Make different folders (or directories) for different classes that you are taking.
- Make different folders for different kinds of content (e.g., data, articles, your writing).

In addition,

- do *not* use the Downloads folder to store anything important, and
- do *not* mix personal content and course content.

Watch one of these to help you understand how your computer is organized. This will save you hours of time and frustration throughout your life.

- [For Windows users \(PCs\)](#)
- [For Mac users](#)
- If you are a Linux user, I am betting that you already know this stuff.

1.2.3 The location of a folder or file: The Path

A “path” is the complete location and name of a file, starting from the hard drive, down to the file or folder of interest. Here are a two examples:

- (Windows 11) C:\Users\Stevenmh\Documents\BI0209W
- (Mac) /Users/stevenmh/courses/209W/

Note they both start big and get more narrow. On Windows, the directory for the harddrive is typically “C:” and on a Mac it is just “/”.

Here are ways to find those paths.

1.2.3.1 File paths in Windows 11 operating system

Open File Explorer, navigate to the file, and click the three-dot menu (...) in the top toolbar to select Copy path. Alternatively, right-click the file and select “Copy as path” (or “Show more options” then “Copy as path”) to paste the full location.

Here are the fastest methods to find a file path on Windows 11:

- Copy as Path (Fastest): Right-click the file in File Explorer and select Copy as path. If you don't see it, select "Show more options" first.
- Top Menu Option: Click the file, then click the three dots (...) in the top menu and select Copy path.
- Address Bar Method: Navigate to the file in File Explorer, click the blank space in the address bar at the top, and it will highlight the full path, which you can copy with Ctrl+C.
- Properties Menu: Right-click the file, select Properties, and look at the "Location" field to see the folder path.

1.2.3.2 File paths on Mac Tahoe operating system

To find a file's path on a Mac, select the file in Finder and press Option-Command-C to copy the path directly. Alternatively, right-click (or Control-click) the file and select "Copy" while holding the Option key to change it to "Copy as Pathname".

Other Methods to Find File Paths:

- Path Bar: In Finder, select View > Show Path Bar to see the file location at the bottom of the window.
- Get Info: Select the file and press Command-I, then look under "Where".
- In the Terminal app: Drag and drop the file into the Terminal window to display its full path.

1.2.4 File types and file name extensions

Here we describe three file types (.R, .csv, and .txt) that you are probably not familiar with. First, we need some background.

Different **file types** allow you to do different sorts of things, and different types are labelled with different **file extensions**. File extensions are little tags at the end of file names that identify the *type* of file it is. A file extension is the little label at the end of a file name that comes after a dot.

Common file extensions include .jpg or .jpeg for many image files, .docx for Microsoft Word documents, and .pages for Apple Pages documents.

Think of a file extension as a label on a container that tells both you and your computer what's inside. Just like you'd label a jar in your fridge as "pickles" or "jam" so you know what to expect, file extensions (those letters after the dot in a filename) tell your computer what kind of information is in that file and which program should open it.

1.2.4.1 Three important file types

- A **.R** file contains your code—the commands you write to analyze data and make graphs. Save your work as “my_Chapter_1.R” so R knows it’s code, not just notes.
- A **.csv** file (comma-separated values) stores your data in rows and columns. This is the standard format for datasets because almost every program—Excel, Numbers, R, Google Sheets, even Word—can read it. When you have data, such as measurements, it’s probably in a .csv file.
- A **.txt** file is plain text with no formatting—useful for notes, project descriptions, or simple lists. Almost any application can open these types of files.

Why it matters: Without the right extension, your computer won’t know what to do with a file. If you save code without .R, R and R RStudio won’t recognize it properly. If data aren’t marked as .csv, R might not read it correctly.

One essential tip: Make extensions visible on your computer. On Mac, go to Finder preferences and check “show all filename extensions.” On Windows, uncheck “hide extensions for known file types” in folder options. You’ll always know what you’re working with.

You will need to be able to see file extensions on your computer.

1.2.4.2 Windows

On Windows 10, you need to make changes to see file extensions by revealing “hidden items”.

Windows 10:

1. Open File Explorer; if you do not have an icon for this in the task bar; click Start, click Windows System, and then File Explorer.

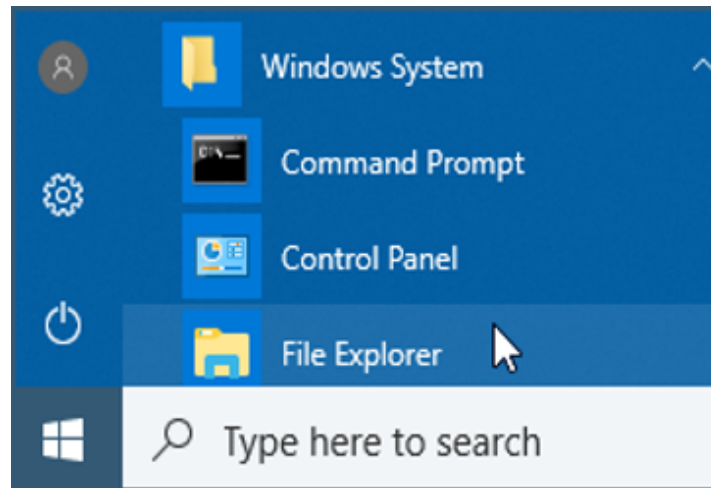


Figure 1.1: Finding File Explorer on Windows.

2. Click the View tab in File Explorer

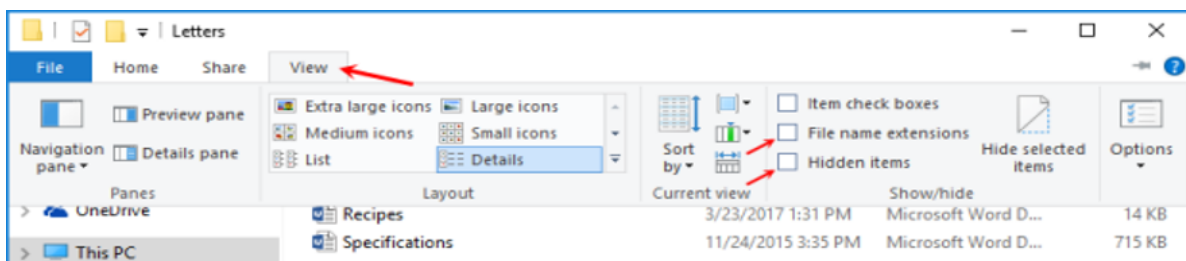


Figure 1.2: View tab in File Explorer

3. Click the box next to File name extensions to see file extensions.

4. Click the box next to Hidden items to see hidden files.

You should now be able to see file name extensions (e.g., .docx, .R) and hidden files.

Windows 11 has a similar, and slightly less complicated layout. In File Explorer, select “Show” and then “File extensions”.

1.2.4.3 Macs

To show all file extensions in macOS (including the latest Tahoe version), open Finder, select Finder > Settings (or Preferences, Command-,) in the top menu bar, click the Advanced tab, and check the box for “Show all filename extensions”.

1.3 Working with CSV files on Windows and Apple computers

It is important that you keep CSV files unchanged. These are our data, and *data are sacrosanct*.

CSV files are easily opened in Microsoft Excel and Apple Numbers, but these applications also encourage you, or force you, to change their format. Below, we explain how to fix that.

Microsoft Excel usually recognizes your CSV file and opens it without any problem. At other times, it will ask you to confirm what kind of file it is, and you will need to specify that values are “delimited” or separated by commas.

When you close a .csv file in Excel, you will be prompted to save it in a different format, as a .xlsx file. Do not do that. Save it as a .csv file.

Apple Numbers will recognize and open your .csv file, similar to Excel. Unfortunately, it always automatically converts it to a different format, .numbers. That means that you have to always **export** to a .csv file type.

When you want to close your file in Numbers - pause! *You must export your file* from Numbers, not merely save it. Do not bother to save it. Just export it. Use the File menu, specifically, File > Export To > CSV, and use whatever name you need to.

1.4 Installing R

Armed with a little knowledge of your computer, you now install R.

1.4.1 What and Why

We will use the Open Source R programming language and environment for doing science, including,

- exploring data,
- testing ideas, and
- modeling dynamical systems like populations of rare and endangered species, disease dynamics, and ecosystems.

You need to download the application to get started.

- R is free.
- R runs on all operating systems.
- R is powerful.

1.4.2 How-to videos (optional)

The following videos might be useful, but they are not required! The next sections describe (in words) how to install R and RStudio.

Optional videos:

1. [Installing R and RStudio](#)
2. [Set up your computer and start to use R via RStudio](#)
3. [Saving figures made with ggplot\(\)](#)
4. [Introduction to ggplot2 \(20 min\)](#)

1.4.3 How-to instructions

1. Find out what version of operating system (OS) your computer is using (e.g., Windows 10 or 11, or Linux Ubuntu, or Mac Sonoma, or other recent editions). This will help determine which version of R to use. If you are using a Mac, this also requires that you determine which version of processor or chip your computer uses (Intel processor vs. Apple M1 chip or higher). Check the processor or chip under “About this Mac” in the *Apple icon menu* in the upper left corner of your screen.
2. Google R (just the letter “R”) to navigate to the *R Project for Statistical Computing*.
3. Find the link in the left Task Bar for *CRAN* (Comprehensive R Archive Network).
4. Find and navigate to a CRAN site near you (e.g., United States or Midwest US or Ohio). These are called *mirrors* because they are copies of each other.
5. Download and install the latest release of R for your operating system (Mac, Windows, or Linux).
6. As of January 2026, it was version 4.5.2. It is a single file to download. You should use whatever the latest version is.
7. In January 2026, it was worked like this:
 - a. Mac - select R-4.2.2.arm64.pkg (for the Apple M1, M2, etc. chip) or R-4.5.2-x86_.pkg (for the older Intel chip)
 - b. Windows - select the “base” package, then “Download R-4.5.2 for Windows”.
 - c. Linux - select your flavor, and follow instructions.

1.4.4 Where

R (and RStudio) will be installed along with all your other applications.

1.4.5 Installing RStudio

RStudio is an integrated development environment (IDE), designed to be used with R. Most importantly for us, it provides a graphical user interface (GUI) that appears identical on Windows, Mac, and Linux computers.

1. Navigate online to the RStudio website.
2. Under Products find RStudio IDE.
3. Select and download the Open Source Edition (RStudio Desktop). You want the desktop version (not the server version).
4. Follow instructions to install on your computer.

1.4.6 R $\neq$ RStudio

RStudio is an interface to R. When we use R in this class, we will use RStudio as our interface, but you need to realize that we are primarily *using R*. We don't need RStudio to do the work, but it makes my job as instructor easier, and makes it a little easier for you as well. So, *when you start to use R, just open RStudio, and it will load R for you*.

As we said at the beginning, you will use the English language to write about ecology, and the R language to explore and analyze ecological data.

1.5 Set up two folders for this class

1. Set up a folder for this class. Call it “BIO209W”. Do not name it “BIO_209W” or “Bio209W” or “ecology”. Just “BIO209W”. It will help us. I don't really care where you put this folder, but you should have a place for all of your classes, and you could put it there. Don't put it in “Downloads”.
2. Next, set up a **subfolder** inside BIO209W called “Rwork”. Do not call it “RWORK” or “R_work” or anything else - just “Rwork”. It will help us. A lot. **Rwork** will be your **working directory**, which is where R looks automatically for data and where it automatically puts output.

1.6 Working in R

To work in R in this class, we will open RStudio. It will open R for you.

1.6.1 Set your “working directory”

In this section, we learn what the “working directory” is and how to set it.

Your **working directory** is where R looks to find stuff by default. R can look anywhere you tell it to, but it looks automatically in your “working directory”.

In this class, you should always use “Rwork” as your working directory.

I can find out where my working directory is by using `getwd()` to **get** my working directory:

```
getwd()
```

```
[1] "/Users/stevenmh/projects/intro_to_ecology"
```

This tells me what R is using as my working directory right now. My current working directory will differ from yours.

You should run this code, and see what you get. What do you get when you run this code (`getwd()`)?

Next, use the “Session” in pulldown menu in RStudio to **Set Working Directory** and then **Choose your working directory** and always choose **Rwork** (which should be inside BIO209W).

1.6.2 Create a *project*

RStudio allows you to organize your work in *projects*. Here we create a new project that we will call **Rwork**.

This requires that you already have a folder called “BIO209W” and a subfolder in that called “Rwork”.

1. In RStudio, use the File menu to select “New Project”, and select “Existing Directory”.
2. In that dialogue, navigate to “Rwork”, and select “Open.”
3. Select “Create Project”.

RStudio will now shift the working directory to your Rwork directory. You can see evidence of this: in the upper right pane which should have Rwork identified in the upper right corner, or in the lower right pane in the Files tab.

From now on in this course, make sure that you are using this project. You can open R by clicking the “Rwork.proj” file in your Rwork directory, or opening RStudio, and selecting the Rwork project in the upper right corner.

1.6.3 Scripts: Start and save a script

Goal: Write a script in R and learn several important programming operations.

After you have prepared your computer as above, you are ready to begin exploring and modeling data. We do that using *scripts*.

A script is a **plain text** file that contains the code and your own comments that you use to address your questions.

The following steps will let you accomplish the goal of this section. The video above showed me opening R with RStudio, and starting a new script.

Writing comments Anything following a hashtag (#) is a comment and R ignores it. You and I will use lots of comments to ask questions of ourselves, and to describe what we are trying to do.

- Use the File menu in RStudio to start a New File, specifically, an R Script. This will open a file in the upper left of RStudio.
- write the following in the new script, and include the hashtags.

```
## [title] e.g., My first R script in BIO 209W  
## [Date]  
## [your name]  
## [the full path to your Rwork directory]
```

Make sure to start each comment with at least one hashtag, #, and remove the square brackets seen above.

ALL OF THE WORK YOU DO MUST BE IN A SCRIPT.

1.6.4 Entering and running code

To enter code, place your cursor on any line in the script, and type, like this.

```
-1:5
```

To run code, simply place your cursor anywhere on a line of code and hit Control-Enter or Control-Return or Command-return (Mac).

Try it with this:

```
-1:5
```

```
[1] -1  0  1  2  3  4  5
```

This will pop up in the Console, which is the lower left pane in RStudio.

What did this create? Do you get what I got above?

1.6.5 Assignment operator

When we create something, we typically want to *assign* it to a labeled object. In this class, we will use an arrow to indicate that we are putting a creation into an object. We create the arrow on our keyboard using shift-comma and a dash ('less than' and a dash)

```
a <- -1:5
```

This puts the series of integers from -1 to 5 into the object `a`.

1.6.6 Examining objects

If we have an object, like `a`, how can we figure out what it is? There are several ways of doing that. The simplest is to “print” it out. This doesn’t mean that we print it on paper. Rather we are printing it to the console, like this:

```
# show or 'print' a  
a
```

```
[1] -1  0  1  2  3  4  5
```

Another way is to look at its “structure”:

```
str(a)
```

```
int [1:7] -1 0 1 2 3 4 5
```

This tells us that this is a vector of integers that is 7 elements long.

Another way is to look in the RStudio *Environment pane* in the upper right hand side of RStudio.

1.6.7 Things you'll find out about R

If you are not used to writing code (and nobody is born that way), here are some things to keep in mind about R and about coding in general.

1. Computers do **exactly** what you tell them. This is both frustrating and also satisfying.
2. R does not know anything or do anything until you **run** the code. Typing or pasting code into a script is just the first step; you also need to **run** the code, with Control-return (Windows or Mac) or Command-return (Mac).
3. A corollary of no. 5 is that you usually need to run lines of code in sequence. For instance, R won't be able to run `str(a)` unless you previously defined `a`, such as `a <- -1:5` (this is because R is an *interpreted* language - R interprets it one line at a time).
4. Case matters in R. R interprets `a` as something completely different than `A`.
5. Spaces *between* things usually don't matter in R. R interprets `2+3` the same as `2 + 3`.
6. Space *within* things usually breaks them, just the way "thin" "gs" is not the same as "things". For example, the pseudo-arrow we use in `f <- 9` is a thing. The compound symbol `<=` means something different than `< -`.
 - i. `<=` is the *assignment operator* and assigns values to objects.
 - ii. `< -` means "less than, minus", where `2 < -3` tests whether 2 is less than minus 3 (it isn't).

Try it.

```
f <- 9
f
```

```
[1] 9
```

Now `f` has the value of 9.

Try this:

```
2 < -3
```

```
[1] FALSE
```

That tested whether 2 is less than -3.

Moving on....

Now let's create a set of 7 uniformly distributed random numbers that are between 0 and 1.

```
b <- runif(7, min=0, max=1)
b
```

```
[1] 0.72010003 0.98954154 0.59849224 0.02597763 0.08130788 0.43052691 0.05781005
```

These will differ every time you do this.

Now we multiply each element in **a** times the corresponding elements in **b**.

```
# multiply a and b
ab <- a * b
## Show a, b, and ab
a
```

```
[1] -1 0 1 2 3 4 5
```

```
b
```

```
[1] 0.72010003 0.98954154 0.59849224 0.02597763 0.08130788 0.43052691 0.05781005
```

```
ab
```

```
[1] -0.72010003 0.00000000 0.59849224 0.05195526 0.24392363 1.72210763
[7] 0.28905024
```

You can confirm at a glance that the first and second elements in **a** times the first and second elements in **b** equal the first and second elements in **ab**.

1.6.8 Combining vectors into a data frame

We will use lots of **data frames** in this course. Think of a data frame as a kind of spreadsheet of data, in which every column has the same number of entries and every row is an observation comprising all of the variables across all the columns.

Here is one way to create a data frame. We assign variables using “=”.

```
d <- data.frame(a=a, b=b, a_times_b = ab)
d
```

	a	b	a_times_b
1	-1	0.72010003	-0.72010003
2	0	0.98954154	0.00000000
3	1	0.59849224	0.59849224
4	2	0.02597763	0.05195526
5	3	0.08130788	0.24392363
6	4	0.43052691	1.72210763
7	5	0.05781005	0.28905024

You could examine the structure of this data frame with `str()`.

```
str(d)
```

```
'data.frame':  7 obs. of  3 variables:
 $ a      : int  -1 0 1 2 3 4 5
 $ b      : num   0.7201 0.9895 0.5985 0.026 0.0813 ...
 $ a_times_b: num  -0.72 0 0.598 0.052 0.244 ...
```

You will also notice now that you can see all of our objects in the Environment pane, in the upper right of RStudio.

You can export data frames from R as well, and programmers call that “writing” a file. The command below will save our data frame as .CSV files that you could open in a spreadsheet application.

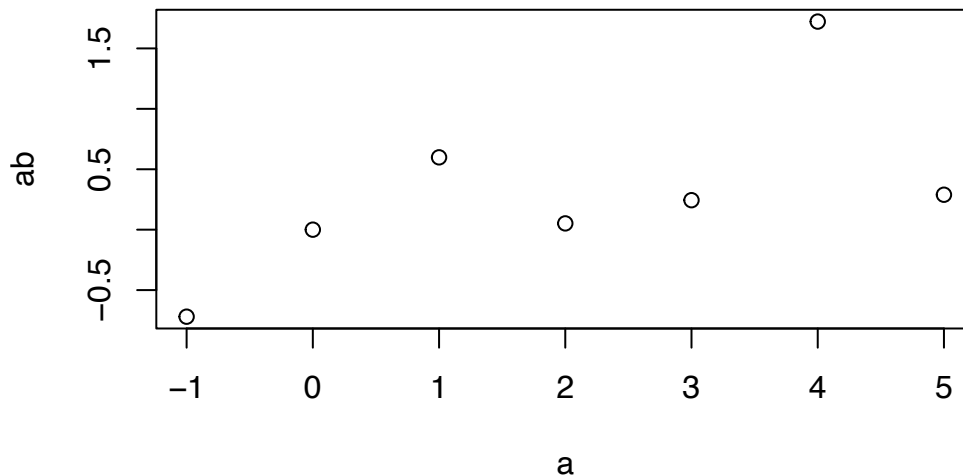
```
# export or write a file
write.csv(x=d, file="myDataframe.csv", row.names=FALSE)
# including row.names=FALSE prevents R from adding row names.
```

This should cause the data frame `d` to get saved to your working directory. Did it work for you?

1.6.9 Plotting data

Now let’s plot some data, including both points and lines connecting them.

```
# type='p' is for points only
plot(a, ab, type='p')
```



R is well known for its graphics capabilities, and we will only just scratch the surface in this course.

1.7 Install an R package

R is open source, and its strength derives in part from the thousands of scientists creating extensions or add-on packages to do particular tasks. By itself, R can do amazing things. With thousands of *contributed packages* (as they are known), R's capabilities have exploded.

You can add a package in a few ways, including.

- Use the Tools menu, select “install packages” and type the name of the package you want.
- use a function, “`install.packages("[packagename]")`”, with quotes but without brackets.

In this class, you will need to install several packages. Let's try installing “ggplot2”.

```
install.packages("ggplot2")
```

This will install ggplot2, and it will also install anything upon which ggplot2 depends. It will install all of its dependencies.

Once you have installed a package, it is on your computer for good. You do not have to install it again, although you may want to update it once in a while to stay current.

Now let's use the plotting functions in the `ggplot2` package.

1.8 Load and use an R package

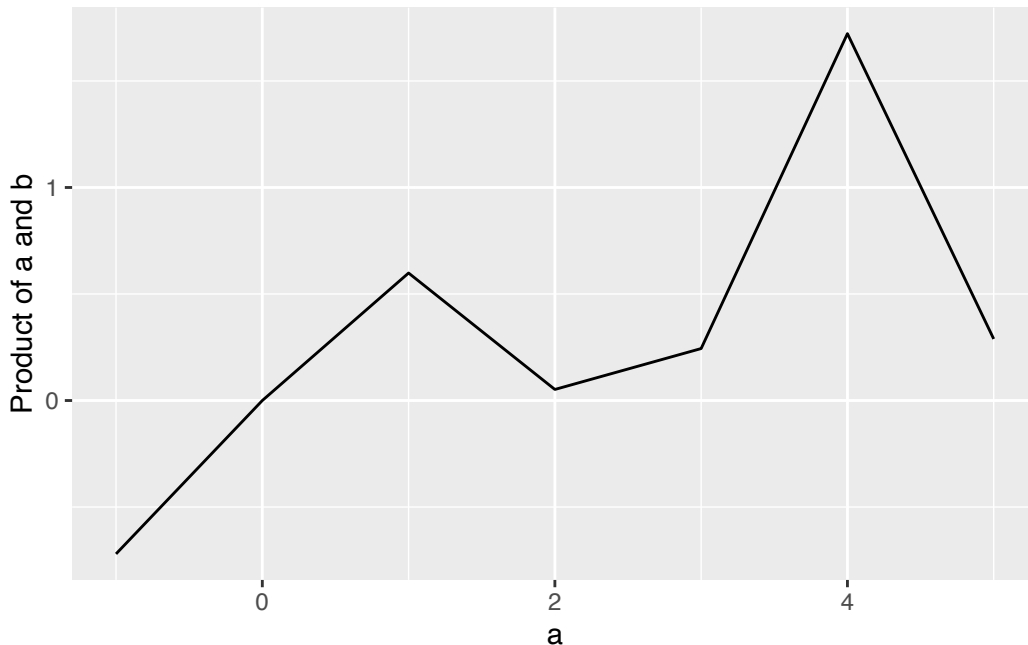
To use a contributed package, we need to *load* it. This simply means waking it up, and putting its code in R's working memory. We do that with `library()`.

```
library(ggplot2)
```

Here we use `ggplot()` and `geom_smooth()` to plot our data, and add a fitted curve.

When we use `ggplot()`, we tell it the name of our data frame and the variables we want to use go inside `aes()`, which stands for “aesthetics”. We also begin to add bells and whistles.

```
## plot data and fit a curve
ggplot(data=d, aes(x=a, y=a_times_b)) +
  geom_line() +
  labs(x="a", y="Product of a and b")
```



Note that with `ggplot`, we use the plus sign (+) to add elements to our graph.

1.8.1 Saving your graphs

We can save this figure as well, in a variety of formats and sizes. By default, these are saved to our working directory, though we can change that if we want to.

```
ggsave("myPlot.png")
```

Saving 5.5 x 3.5 in image

```
ggsave("myPlot.jpg", width=7, height = 3)  
ggsave("myPlot.pdf", width=3, height = 5)
```

1.9 Getting help

R has a bit of a learning curve. What should you do if you get stuck? It all depends what sort of problem it is.

1. First, ask a classmate.
2. Use the Help tab in the lower right pane in RStudio. (Similarly, type `?mean` in the Console to get help on the function `mean()`.)
3. Ask another classmate.
4. Ask me.
5. Search online.
6. Did I mention you could ask me?

1.10 Deliverables

For this assignment, upload to the LMS three files that you created:

1. `myPlot.png`
2. `myDataframe.csv`
3. the R script, into which you wrote all your code.
 - a. This file must be written in R/RStudio, and the name must end with the file name extension, “.R”, such as “Stevens_script_1.R”,
 - b. comments, including at least a title, the date, your name, and the path to your Rwork folder.

Congratulations. *C'est fini!*

2 Environmental Justice: How income relates to environment.

2.1 Background

The types of places that organisms live—their habitats—depends on a variety of factors. *Habitat selection*, the selection of particular habitats, results from real and perceived tradeoffs between the benefits from foraging and reproducing versus perceived risks of mortality (Stephens and Krebs 1986; Morin 1999).

In addition to selecting their habitats, organisms also construct elements of their own habitats, including nests and burrows, as well as broader habitat alteration. This is known as *niche construction* (Odling-Smee and Erwin 2013). Beavers build dens and dams, birds and wasps build nests, and bacteria, ants, and humans all construct elaborate multi-layered cities of thousands or millions of individuals. At the same time, organisms often damage the very habitats they create, through over-consumption of resources and over-production of waste products.

We know a great deal about how our own species (*H. sapiens*) construct elements of their own habitats. We also know a lot about how we migrate and disperse from place to place. Nonetheless, we have a lot to learn about our own species, and, in particular, about which factors determine our preferred habitats and determine the changes we make to those habitats.

Economic wealth seems to influence many of the choices by *H. sapiens*, including our preferred habitats. Ed Wilson (1984), a famous evolutionary biologist, has proposed the *Biophilia hypothesis*, that humans love nature. Biomedical researchers have shown that we are happier and heal from injury and illness faster when we have access to plants and to the outdoors (Franklin 2012). If wealth generally gives us greater access to what we want, will wealthier people more often put themselves in closer connection to nature?

Specific socioeconomic data about *H. sapiens* populations may be particularly useful understanding both the causes and consequences of relations between wealth and nature. Wealth can be measured in a variety of ways. Median income, and percent of the population below the poverty line are common ways to measure the average wealth of individuals or small family groups.

Specific data about our landscapes may be particularly useful in generating ecological understanding of *H. sapiens*. Tree cover is the two-dimensional coverage of tree canopies that you would perceive in an aerial photo in seasons when trees have their leaves. Tree cover is relatively easy to measure from satellite imagery. Temperature, or the heat experienced by residents, can be a serious problem in urban areas because they create heat “islands”. Urban heat islands are areas which tend to absorb more solar radiation than would a natural landscape and then re-radiate this as heat experienced by its human and non-human residents. During heat waves, residents in urban areas who are especially vulnerable can suffer severe heat exhaustion or death. Some urban areas tend to get much hotter than others, and it is often related to tree cover and green space generally.

Trees provide enormous ecosystem service benefits, including to help mitigate pollution via uptake of carbon dioxide and carbon monoxide (CO₂ and CO) and other greenhouse gases. Trees can also help reduce pollutants that cause direct negative effects on human health, such as fine and ultrafine particulate matter (PM_{2.5}, PM_{0.1}).

In this research project, we will use freely available online data to address whether there is an association between wealth and the wild. Specifically, we will address whether, if we compare different geographic locations, will we see a relationship between the median income of an area, and the percent cover of trees? The data we collect will allow us to explore a variety of related ideas, but we will start with testing Wilson’s *biophilia* hypothesis, which predicts that humans love nature. We will test my corollary of this, that humans who have the greatest degree of freedom of choice (the wealthy) will tend to live in areas with the greatest connection to Nature (a lot of trees).

View your mission here: <https://youtu.be/BABM3EUo990>
or here:

2.2 Methods

Work in pairs. Share introductions with someone and get to work.

We will collect data on tree cover (%), heat, and wealth using two different platforms:

1. [Tree Equity Score website](#)
2. OurTrees application in *i-Tree*

The first of these two platforms, Tree Equity Score, gives fairly detailed information on cities over 50,000 residents in size. It provides information related to environmental justice, including levels of poverty, tree cover, and heat. Our second data source is from *i-Tree.org* and it provides fairly detailed information about tree cover and the ecosystem services they provide. (Ecosystem services are features of ecosystems that benefit humans, such as storm water protection, and shade for buildings). *i-Tree* also provides median household income.

2.2.1 Instructions

Tree Equity Score data

1. Go to our class [Google Sheet for Tree Equity Score cities](#), and enter the name of a city (pop. > 50,000) that you will investigate. It must be different than what anyone else has selected. Add your name to your selected city.
2. Go to the [Tree Equity Score website](#).
3. Search for your city.
4. Find the data for the city: click on the map *inside the area of your city*. Note that a map will typically display more than just your city. It will also display adjacent regions. For instance, the map for “Cincinnati, OH” displays Hamilton County and Covington, KY. Selecting an area will cause a data summary window to pop up on the left side. **Confirm** that when you click on an area of the map, it is **inside the city you intended**.
5. Selecting an area link to a **Municipal Report**. This report provides information for the city as a whole. **Confirm** that when you click on an area of the map, it is **inside the city you intended**. Select the link, Municipal Report.
6. Pick the data most relevant to our question: adjust the setting for the right-hand column graph to show *Tree Cover (%) vs. People in Poverty %*. You will use this column graph in addressing our overall question.
7. The Municipal Report provides a Tree Equity Score, urbanized area population estimate, percent seniors, and a percentage of people in poverty. Enter these into the Google Sheet. Add one Miami uniqueID per group to each observation (row).
8. Save the web page as a PDF, and name it with the city name and “TES” (e.g., “Cincinnati_TES.pdf”). If you would like assistance saving the PDF, read the instructions under the button “Share report ->”. Make sure to close the instructions before you save the page. This file will be a record of your data.
9. Make sure that you have entered the Tree Equity score and poverty percentage in the Google Sheet, and that you saved to your own computer your column graph of tree cover vs. percent of people in poverty.

OurTrees app, in *i-Tree*

1. **OurTrees application** in *i-Tree*. Select a location in which you are interested. Start with any location name, and select “Search”. The app will show you a location. If you would like to, you can change the location by simply clicking somewhere else on the map. You can choose either Street view or Satellite view (the data will be the same).
2. Enter the name of your selected location into our class a new spreadsheet, [Google Sheet for OurTrees](#).
3. In the OurTrees app, select “Get Results! ->”.
4. From the “Benefits” tab, collect data on tree canopy cover (%), annual uptake of CO₂ equivalents and 2.5 micron particulate matter (PM_{2.5}). Enter these into the OurTrees Google Sheet.

5. From the “Community” tab, collect data on total population size, number of seniors (>64), percent minority, median annual income, and percent impoverished. Enter these into the OurTrees Google Sheet.
6. Repeat the above steps for *nine more* locations anywhere in the world. Your locations must be different than what anyone else has selected in the OurTrees spreadsheet. Add one Miami uniqueID per group to each observation (row).

2.2.2 Graphing data to assess our predictions

I argued that the Biophilia Hypothesis leads to the prediction that tree cover will correlate positively with wealth. To evaluate this prediction, we should make a graph in which wealth is on the x-axis (horizontal axis) and tree cover is y-axis (vertical axis).

A couple of minor points: (1) Our prediction also implies that we should see a negative relation between poverty and tree cover. (2) We typically assume that the variable on the y-axis is a response to the variable on the x-axis, that is, that the x variable is causing variation in the y variable.

Given our data are *continuous variables* rather than categories, we should start by making a *scatterplot* of median income (x) vs. tree cover (y).

As the semester progresses, you will find R is very useful. Below, I provide you with code for an R script to easily make this graph.

```
# if needed, install these packages
# install.packages("googlesheets4")
# install.packages("ggplot2")

# load these packages
library(googlesheets4)
library(ggplot2)

# Import data.
# This code will require that you have permission to access our G-Sheet.
# You may be asked to select various options and/or sign in to your
# Miami Google account.

# The function read_sheet() requires a Google Sheet URL in quotes, and
# allows you to skip lines that are not data,
# such as a table description.

d <- read_sheet("https://docs.google.com/spreadsheets/d/1uYCuTziJiLNvVOQ9Q65BxdWcNgDF6EnF8L1"
                skip=2)
```

x Request 1 failed [429: RESOURCE_EXHAUSTED].

i Will retry in 5s.

\ Retry happens in 4s

| Retry happens in 3s

/ Retry happens in 0s

x Request 2 failed [429: RESOURCE_EXHAUSTED].

/ Retry happens in 0s

i Will retry in 7.1s.

/ Retry happens in 0s

/ Retry happens in 0s

\ Retry happens in 5s

| Retry happens in 2s

v Request 3 successful!

| Retry happens in 2s

| Retry happens in 0s

v Reading from "iTree_OurTrees".

v Range '3:10000000'.

```
# Uncomment to show us the names of the variables
# names(d)
```

```
# make a scatterplot
```

```
plot1 <- ggplot(data=d, aes(x=Median_Income,
                             y=Tree_canopy_cover,
                             label=`City (only)`)) + geom_text() +
  labs(y="Tree cover (%)", x="Median income", title="No relation twixt tree cover and income")
```

```
ggsave("myCoverIncome_text.png", plot=plot1, width=10, height = 10)
```

```
plot2 <- ggplot(data=d, aes(x=Median_Income,
                             y=Tree_canopy_cover,
                             label=`City (only)`)) +
  geom_point() +
  labs(y="Tree cover (%)", x="Median income", title="No relation twixt tree cover and income")
```

```
# save the figure (dimensions in inches)
ggsave("myCoverIncome_points.png", plot=plot2, width=7, height = 7)
```

Here is a subset of data I collected on the places where members of my family lives.

```
d10 <- d[1:10,]
myPlot <- ggplot(data=d10, aes(x=Median_Income,
                               y=Tree_canopy_cover,
                               label=`City (only)`)) + geom_text() +
  labs(y="Tree cover (%)", x="Median income", title="No relation twixt tree cover and income")
myPlot
```

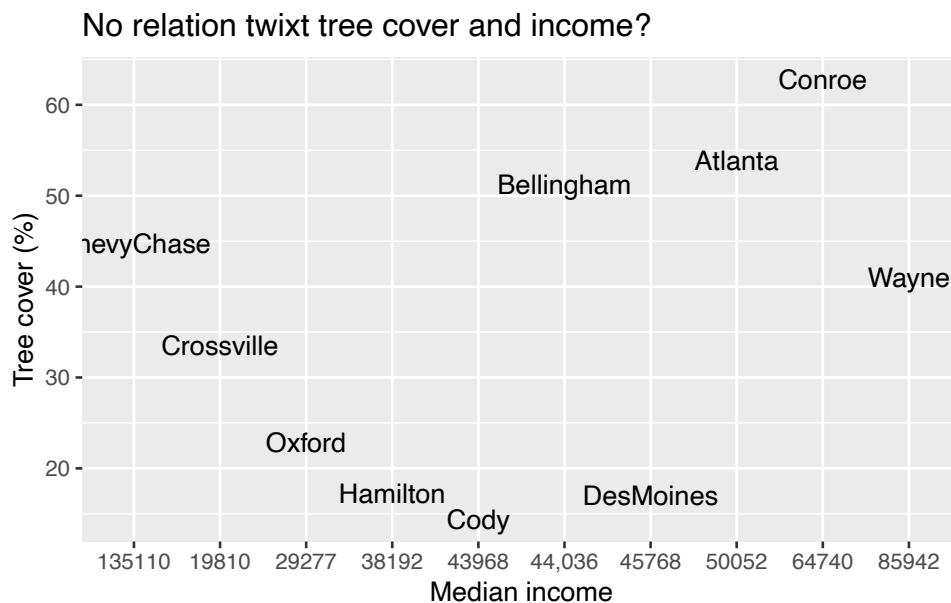


Figure 2.1: The noisy but perhaps positive relation between percent of an location covered in trees and the median income of that location.

Refer to the scatterplot of my data (Figure 2.1). What do we find? Discuss the following questions with your colleagues, and propose and explain some possible answers.

1. Is there a relationship that is consistent with our hypothesis? Does tree cover increase with income?
2. Notice the cities that I selected. What else do you know about these locations, aside from income, that might influence tree cover?

3. What else might be going on that helps create this relation or might obscure a stronger relation?
4. After looking at these data, would you want to select locations in a different way?

2.3 Deliverables

Results and discussion

- In one document, put two figures side-by-side:
 - i. your bargraph of % tree cover vs. % people in poverty (copied from the Tree Equity site) side-by-side, and
 - ii. your scatterplot of tree cover vs. median income that you made using everyone's data, based on the iTree/OurTrees website that you shared in the Google spreadsheet.
 - iii. include a figure legend that states what the graphs are in a way that is intelligible to a naive reader.
- Provide a brief explanation of your findings, including
 - i. the results described in a factual manner, without interpretation,
 - ii. an explanation of whether the results are consistent with the biophilia hypothesis, and
 - iii. a new question that emerges due to the inconsistencies between the data and the hypothesis; explain why this question is a logical extension of your current results and interpretation.

3 Physical Environment: A Fate of Earth

...in which we reveal part of our climate reality.

3.1 Goals

In this assignment, you will

- Learn about the history of our perspective on global warming.
- Use R to display estimates of global temperature anomalies.
- Describe the current temperature trajectory.
- Compare and contrast changes in temperature between humans and Earth.
- Background reading: Hansen et al. (2012). Perception of climate change. *Proceedings of the National Academy of Sciences*, U.S.A.

3.2 Background

The year that you were born was much warmer than usual—and it won't ever be that *cool* again in your lifetime.

Earth is getting warmer due to human activities, it will push it well beyond what earth has experienced in hundreds of thousands of years. How and why Earth is warming has been known for many decades. In the early 1980's, even Exxon's scientists accurately predicted the global warming that has occurred because we failed to act to change our behavior.¹

In this exercise, you'll graph data from the National Centers for Environmental Information, part of NOAA (US Dept of Commerce). These temperatures are anomalies from a long term mean. The reference they use is the period from 1951-1980.

Using the R script, you will construct a 95% confidence interval on expected mean surface anomalies. A "95% confidence interval of a mean" is an interval, if so constructed, that would tend to include the true mean about 95% of the time, given many, many repeated experiments conducted under identical conditions. It is based on frequentist statistical methods. It is

¹Exxon suppressed these internal reports, and they became available only due to a lawsuit.

actually pretty difficult to interpret, but it is a standard measure of uncertainty. You will construct such an interval for past and future global surface temperature anomalies.

Overthinking: *Frequentist statistical methods (those that you will all attempt to learn in intro stats class) give you a P -value, which is the probability of your data or more extreme data, given that your null hypothesis is true (i.e., your model is correct). That's weird. The data are real - they don't need a probability. A 95% confidence interval is an interval that we construct in a way where we would capture a "true" effect 95% of the time we repeat the same experiment under the same conditions. In contrast, Bayesian statistics would give us what we actually want: the probability that our model or hypothesis is correct, given our data. That is called conditional probability and is what a Bayesian analysis would give you. We might do that later in the course.*

3.3 Activity

In this activity, you will download and import data, and then make a forecast about future temperatures. That forecast will include an *expected* value and a measure of uncertainty. The *expected* value is the most likely value, but we also acknowledge that any time we estimate a most-likely value that there is always uncertainty.

Below, we set up our work environment, load data, work with data, and then save deliverables. This work flow will be natural after a while.

3.3.1 Set up

First, if you have not already done so, create a script file that you will call `global_anomalies_YourUniqueID.R`.

After starting a script, we always select the appropriate project and *working directory*. I do this by...

...glancing at the upper right corner:

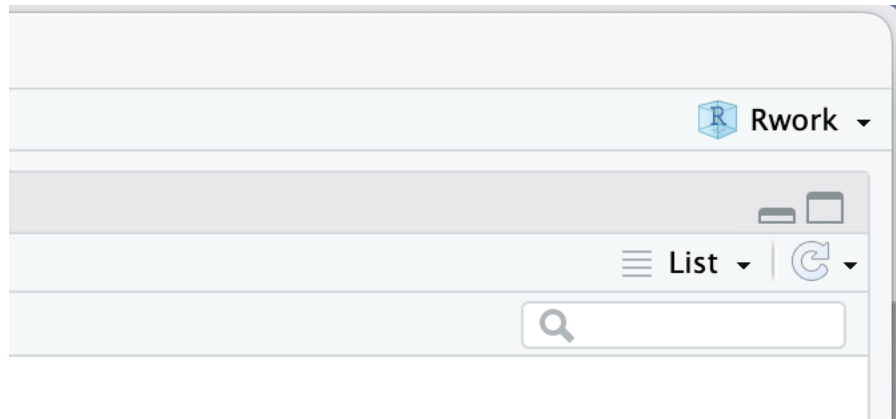


Figure 3.1: RStudio shows which project you are using in the upper-right corner

...and by looking in the File tab in the lower right pane - that should be your working directory, which should be Rwork.

The next thing we do is **install** and then **load** packages that we know we will need. Remember, you only need to *install* a package onto your computer *once*. We **load** contributed packages in each session that we need them. In this course, you should always load **tidyverse**, at a minimum.

First we install.

```
## install tidyverse
## tidyverse is a "universe" or collection of related R packages
## that have a particular coding style that helps a lot.
install.packages("tidyverse", repos = "https://cloud.r-project.org")
```

Remember, we only have to do that *once*, and then it is sitting on our computer.

However, anytime we start a new session in R, we have to *load* these packages in order to use them. We do that here.

```
## load R packages
# this makes these packages available to R in this session.
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v ggplot2    4.0.2      v tibble     3.3.0
```

```

v lubridate 1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become

```

3.3.2 Import data

1. Go to our [class data repository](#), and download `global_surface_anomalies_2025.csv`.
2. Make sure the file remains in the same simple format, “comma separated values” (.csv).
3. Put `global_surface_anomalies_2025.csv` in your working directory, `Rwork`.
4. Open `global_surface_anomalies_2025.csv` in a spreadsheet application. Don’t save the file in the spreadsheet application. Do **not** save it in the spreadsheet application because it may try to change the format.
5. With the file open, learn what is on
 - a. the first three lines
 - b. the fourth line
 - c. the remaining lines.
6. Do **not** save the file in the spreadsheet application. Just close and don’t save it - it will still be there, unchanged.
7. Continue to make sure the file remains in the same simple format, “comma separated values” (.csv).

We import or “read in” the data so that R can work on it. Here we put it in an object called `temps`. We skip lines in the file

```

# read in data
temps <- read.csv(file = "global_surface_anomalies_2025.csv", # full file name
                  skip=3, # skip these lines because they contain metadata.
                  header = TRUE # Tell R that the first row is column names
                  ) # last parenthesis

```

3.3.3 Finding out information about the temperature anomalies

Next we take a peek at the data with a summary.

```
summary(temps)
```

	year	deg_C
Min.	:1880	Min. :-0.48000
1st Qu.	:1916	1st Qu.: -0.20000
Median	:1952	Median : -0.04500
Mean	:1952	Mean : 0.06771
3rd Qu.	:1987	3rd Qu.: 0.28000
Max.	:2023	Max. : 1.17000

This shows us the minimum, maximum and mean of each numeric variable.

It also shows us the 1st, 2nd, and 3rd [quartiles](#). The median is the 2nd quartile. Quartiles are three divisions in ranked data, such they divide the data into four equal parts, or quarters. The 1st quartile is the value that divides the lowest 25% of the values from the larger values. The 2nd quartile (i.e.) is the value that divides the lower 50% from the upper 50% of the observations. The 3rd quartile is the value that divides the highest 25% of the observations from the lower values.

3.3.4 Minimum and maximum

I'd like to know *when* the minimum and the maximum occurred. To find this out, I'll use a two step process: I will find the min and the max, and then I will use those to take a *subset* of the data frame where the anomaly is equal to either the minimum or equal to the maximum.

First, I find the minimum and the maximum.

```
## get the min and max
min.anomaly <- min(temps$deg_C)
max.anomaly <- max(temps$deg_C)
```

Now, I will use these to take a *subset* of the data frame where the anomaly is equal to either the minimum or equal to the maximum. The “OR” operator is the vertical line that programmers call a *pipe*. On your keyboard, it is above the return or enter key.

```
## use the min and the max
# note the two equal signs, a pipe, and then two more equals signs.
subset(temps, deg_C==min.anomaly | deg_C==max.anomaly )
```

	year	deg_C
30	1909	-0.48
144	2023	1.17

These show us all of the entries in our data that the anomaly is equal to the minimum OR equal to the maximum.

3.3.5 Using a year

Here I find out the average temperature anomaly in the year I was born. I want YOU to find out the anomaly in the year *you were born*.

```
## Find the anomaly in the year you were born
subset(temps, year==1960) # use your year with two equal signs
```

```
  year deg_C
81 1960 -0.03
```

In my birth year, it was just a touch below the 1951-1980 average.

3.3.6 Plotting information

Here we are going to start visually examining the anomalies.

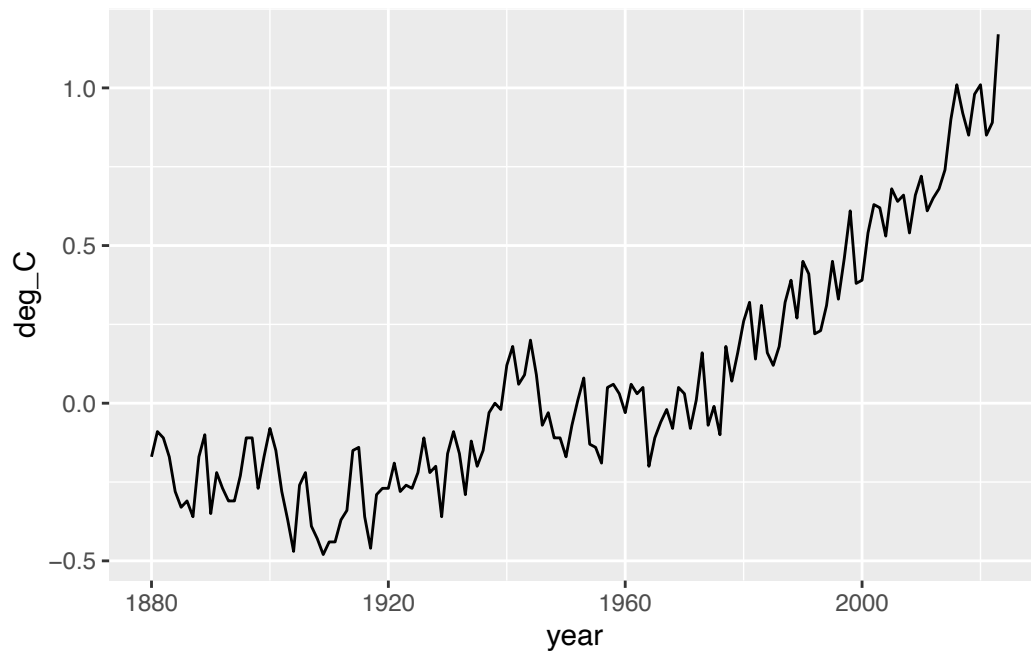
We will create a linegraph using `ggplot()`. Our first step will create the graph, but it won't display it right away. I include comments on each line to remind you what everything does.

```
## we will create a plot object and stick it in "p"
p <- ggplot(data=temps, # the data frame to use
  ## aes() stand for "aesthetics"
  aes(x=year, y=deg_C), # define x and y variables to plot
) + # closing parenthesis and add a plus sign to add another graphical element
  geom_line() # the type of graph to plot using x and y
```

If you do this correctly it will seem as though nothing happened. If you did something wrong, you might get an error message in the console. If R can't find the function "ggplot", that is because you didn't load `tidyverse` (or `ggplot2` which is a subset of `tidyverse`). Another common error is an incomplete statement that might come from a missing parenthesis or something. In that case, R might think that you are not finished making your statement and it will keep adding plus signs so that you can finish your coding statement.

Now we display the graph.

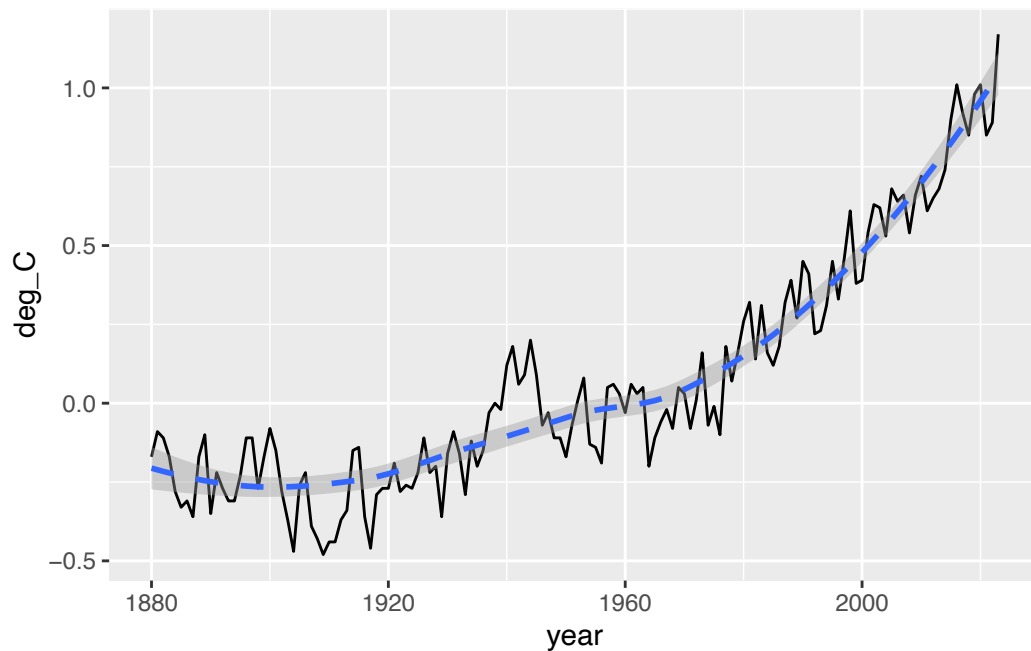
```
p
```



Here we add a trend line using a fancy statistical smoothing technique. By default, it will include a grey uncertainty region.

```
## add a trend line  
p + geom_smooth(linetype="dashed")
```

``geom_smooth()`` using `method = 'loess'` and `formula = 'y ~ x'`



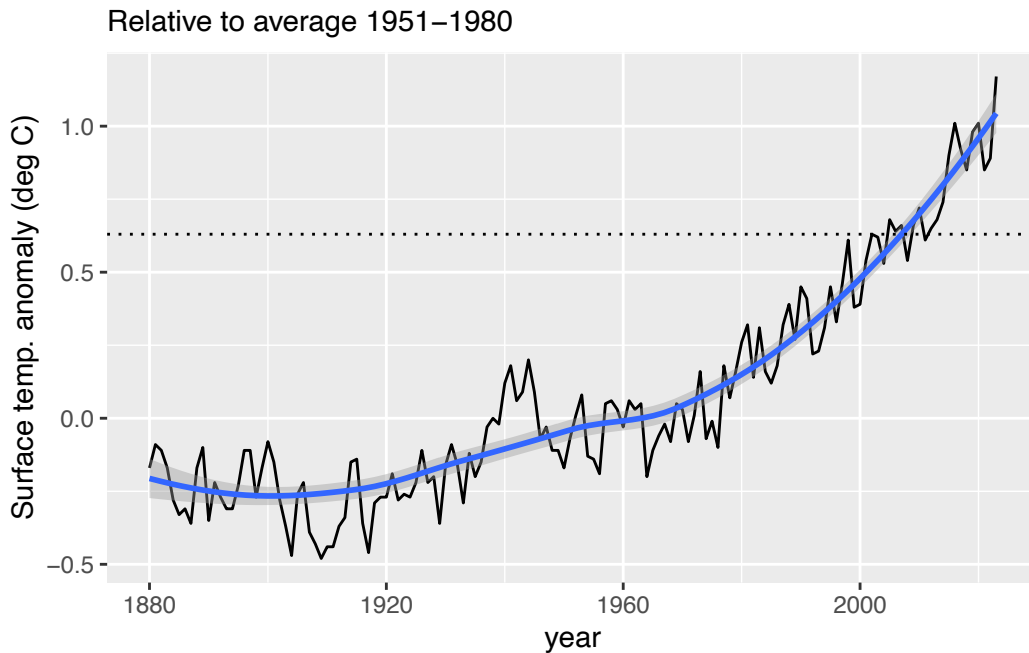
Here we add prettier, more descriptive labels. Add a line for the anomaly in 2002.

```
# first find the anomaly in 2002
anomaly.2002 <- temps$deg_C[temps$year==2002]
anomaly.2002
```

```
[1] 0.63
```

```
p + geom_smooth() +
  labs(
    y="Surface temp. anomaly (deg C)",
    subtitle="Relative to average 1951-1980"
  ) +
  geom_hline(yintercept = anomaly.2002, linetype=3)
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



3.4 Predicting future anomalies

Major goals of science are understanding and prediction. We could spend a long time discussing the connection between these two, but from now, we just want to forecast what our future might look like.

The statistical details of this are way beyond this course, and they are not even the way you would do it if you were serious about. Nonetheless, it won't be too far off, and the intuitive part applies generally to making any set of predictions. We will merely ask the computer to do some math to accomplish the following:

1. create an educated guess for the temperature trend. This will include,
 - a. uncertainty that comes from how much the anomalies bounce up and down from year to year, and
 - b. a trend that comes from averaging nearby values.
 - c. information that nearby years may be more similar to each other than years that are really far apart.
2. given the observed uncertainty and the trend, make an educated guess for future observations, taking into account that *each future year adds uncertainty that we can only guess at.*
3. Make a figure showing the future forecast.

So, let's start with step one. and create a statistical model of the observations. We start by loading another package.

```
## Load a necessary R package
library(mgcv) # stands for "Mixed Generalized Additive Model Computation Vehicle"
```

Loading required package: nlme

Attaching package: 'nlme'

The following object is masked from 'package:dplyr':

collapse

This is mgcv 1.9-3. For overview type 'help("mgcv-package")'.

Now we ask R to fit the model. Don't worry about the details of the code. You can ask me if you are interested.

```
## Create the model
mod <- gamm(deg_C ~ s(year), # "deg_C as a function of year
            data=temps, # data frame
            correlation = corCAR1(form=~year), # specify autocorrelation among samples
            method="REML" # fit using restricted maximum likelihood
            )
```

Next we have to decide what years we want predictions for. Let's go for 50 more years, to 2072

```
## years for fit and prediction
years = data.frame(year=1880:2072)
```

Now we apply the model to past and future years and make a data frame and keep the uncertainty estimates it generates. We also combine our specified years with the model.

```
## include uncertainty with se.fit = TRUE
preds <- data.frame(
  predict(mod$gam, newdata=years, type="response", se.fit=TRUE)
)

# combine columns of data frames
my.predictions <- cbind(years, preds)
```

Let's look at the first three rows of what we've made.

```
# first three lines , and all columns
my.predictions[1:3, ]
```

	year	fit	se.fit
1	1880	-0.1871517	0.06106168
2	1881	-0.1918986	0.05677760
3	1882	-0.1966483	0.05272457

This shows that the first column is the year, the second (`fit`) is the expected local estimate, and the third (`se.fit`) is the estimate of the uncertainty, which is the standard error of the expected value.

Now we'll create 95% confidence intervals for each year. Confidence intervals are intervals which would contain the “true” value (if there is one) a specified proportion of times, if you repeated the experiment a really, really large number of times. Ninety-five percent confidence intervals exclude the lowest 2.5% and the upper 97.5% regions.

```
### Create 95% confidence intervals
crit.t.low <- qt(0.025, df = df.residual(mod$gam))
crit.t.high <- qt(0.975, df = df.residual(mod$gam))

### this requires that you loaded the tidyverse package
my.predictions <- my.predictions %>% # my data
  ## transform the data set to include two new columns
  transform(
    upper.conf.lim=fit + (crit.t.high * se.fit),
    lower.conf.lim=fit + (crit.t.low * se.fit))
###
## view the first three rows
my.predictions[1:3, ]
```

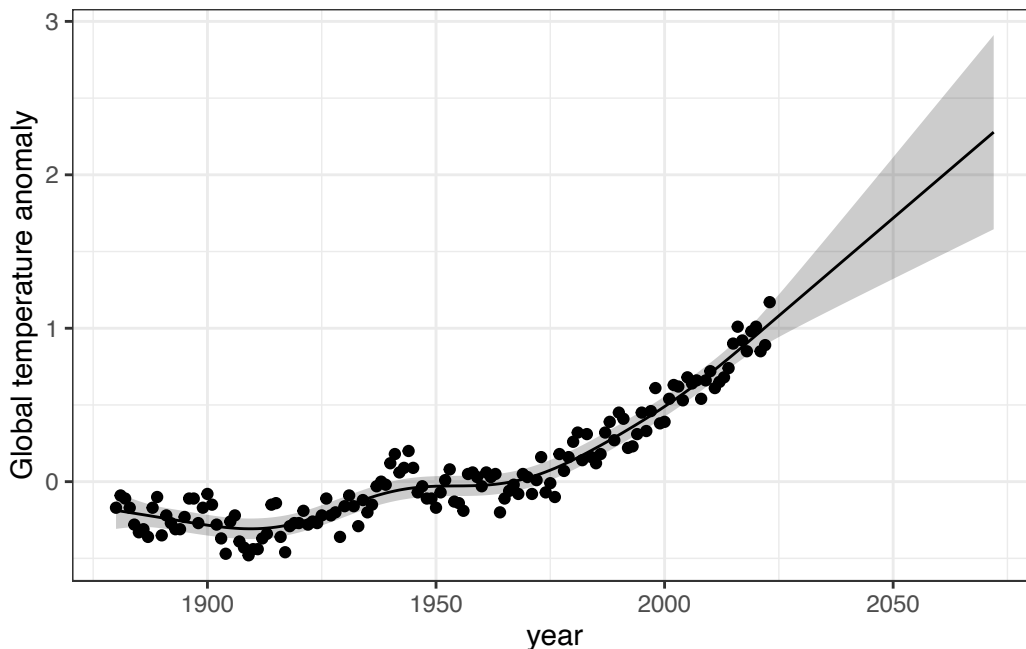
	year	fit	se.fit	upper.conf.lim	lower.conf.lim
1	1880	-0.1871517	0.06106168	-0.06640801	-0.3078953
2	1881	-0.1918986	0.05677760	-0.07962626	-0.3041709
3	1882	-0.1966483	0.05272457	-0.09239046	-0.3009061

Now we see we've added upper and lower confidence intervals. Next we'll create a graph of our predictions, using several particular graphical features.

```
## basic data and x, y values
g <- ggplot(data = my.predictions, aes(x = year, y = fit)) +
  ## adding a wide region or band (a ribbon)
  geom_ribbon(aes(ymin = lower.conf.lim, ymax = upper.conf.lim,
                 x = year),
            alpha = 0.2, fill = "black") +
  ## add the expected local trend
  geom_line() +
  ## adding the points for each year
  geom_point(data = temps, mapping = aes(x = year, y = deg_C),
            inherit.aes = FALSE) +
  ## add a better label for the y axis
  labs(y = "Global temp. anomaly")
  ## simplify the image
```

Now we show the graph we created.

g



The grey area represents the 95% confidence interval for the trend observed years, and also for the trend in the future years. It gets wider because each year's uncertainty compounds with the next year's uncertainty. [Your figure will look a little different the this one.]

Finally, we save your figure. This is the graph you should use for your assignment.

```
## Save a PNG file to your working directory
## THIS IS THE GRAPH TO INSERT INTO YOUR ASSIGNMENT
ggsave("Global_temps.png", # name for your figure
       plot=g, ## identify which graph you want to save,
       height=5, width=5)
```

3.5 Does Earth have a fever?

In this last section, we ask whether Earth's temperature may be a little too high by comparing it to a human who has a fever.

First we calculate the percent increase in temperature for Earth over the past 70 years and a human experiencing a serious fever.

Average temperature for humans is 37 C, and a serious fever is considered 39 C, so an *increase* of 2 C.

Earth's 20th century average temperature is 13.9 C. Let's calculate the increase since 1950.

```
## First we take a subset of our data set.
a <- subset(temps, year==1950)
b <- subset(temps, year==2022)
increase <- b$deg_C - a$deg_C
increase
```

```
[1] 1.06
```

Next we calculate these increases (fever and global warming) as *percentage change*.

```
## percent increase for Earth
perc.Earth <- increase / 13.9 * 100

## percent increase for humans
perc.fever <- 2/37 * 100

perc.Earth; perc.fever
```

```
[1] 7.625899
```

```
[1] 5.405405
```

So, what do you think? Does Earth have a fever? In what ways do you think comparing temperature variation of Earth to that of a human is or is not a useful thing to do.

3.6 Questions to answer

1. What is a temperature anomaly?
2. In the file “global_surface_anomalies.csv”, what is on
 - a. the first three lines?
 - b. the fourth line?
 - c. the rest of the data?
3. What are the minimum, median, and maximum anomalies?
4. When (in what year) were the minimum and maximum anomalies?
5. What was the global average surface temperature anomaly in the year you were born? Provide the year and the anomaly.
6. What does the grey band in the figure represent, and why does it get wider in future years?
7. Explain whether you think Earth currently does or does not have a fever, and whether comparing temperature variation of Earth to that of a human is or is not a useful thing to do.

3.7 Deliverables

Turn these into the relevant Canvas assignment before the due date.

1. First, put all your answers, including your graph with the confidence interval, into a .doc or .docx file and make fit on a single page. Turn in your answers and your graph to the relevant Canvas assignment Make sure that your name is on the file in the header!
2. Second, turn in your R script, named `global_temps_YourUniqueID.R` as a second file. This must NOT be a Word document. It must be a .R file.

4 Individual responses to the environment: What makes lianas relatively successful in stressful habitats?



Figure 4.1: Lianas are woody tropical vines, and can be really common in some tropical forests.

4.1 Goals

In this assignment, you will

- Investigate reasons for the unreasonable success of lianas.
- Develop and use your understanding of plant growth and responses to water deficits to help make predictions about the relative success of lianas and trees.
- Evaluate scientific hypotheses and predictions by (i) enhancing your data visualization skills using `ggplot()`, and (ii) making graphs of your own design.
- Background reading:

4.2 Background

- Download Smith-Martin et al. (2019), which is available in [our PDF archive](#). Read the introduction to help you understand the context of this chapter.

The term “liana” refers to tropical woody vines, which can be very common and can account for as much as 25% of the woody plant biodiversity in some tropical forests. They achieve some of their highest abundance in seasonal subtropical forests Figure 4.2. These are regions with distinct wet and dry seasons; why are lianas successful in these seasonal subtropical forests?

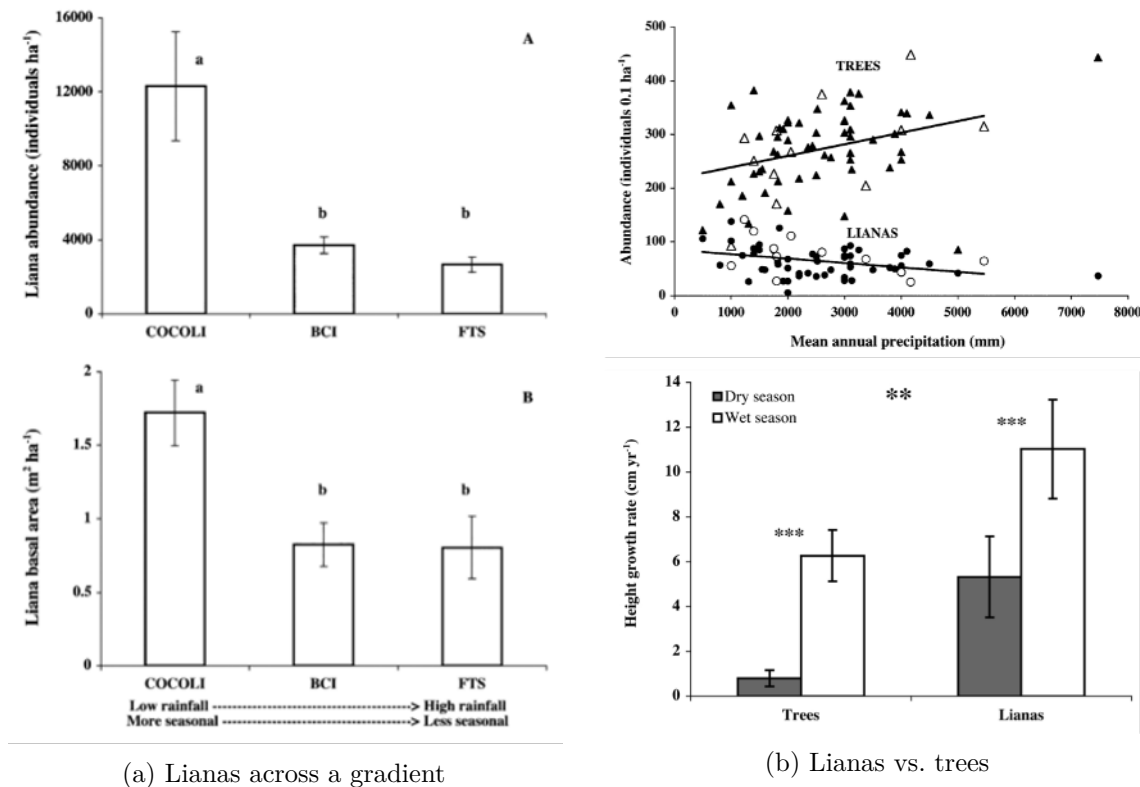


Figure 4.2: Lianas increase in abundance and grow better than trees in the the dry season in seasonal subtropical forests. Figures 1-3 from Schnitzer (2005).

One possible explanation for lianas' success in seasonal forests might have to do with their physiology and architecture (Schnitzer 2005). Lianas may possess an unusual ability to regulate their water potential (Smith-Martin et al. 2019). In order for plants to grow and reproduce, they undergo photosynthesis, and maintaining a less negative water potential may help lianas perform better than trees in seasonal environments. (See video lectures on soils, plant photosynthesis, water potential, and heat regulation). This ability may come from the depth at which their roots grow and forage for water, their very large vessels (xylem) that carry water up through the stems, or it could be something else.

Christine Smith-Martin and her colleagues (2019, Figure 4.3) **hypothesized that lianas can photosynthesize and grow better than trees during the dry season because lianas can maintain a less negative water potential than trees**. To test their ideas, they set up a common garden experiment with, six liana and six tree species in replicate plots, in region of Panama with highly seasonal precipitation (Figure 4.3). They watered half of the plots, and measured a wide variety of variables including photosynthetic rate, pre-dawn and midday leaf water potential, water use efficiency, many other variables about lianas and trees.



Figure 4.3: Geographic location of study sites in Smith-Martin et al. (2019). Inset, Chris Smith-Martin, seen here in a tree crane. There is a strong seasonality gradient from the Atlantic to Pacific coasts of Panama.

You will evaluate the hypothesis proposed by Smith-Martin and her colleagues using their publicly available data. These data are also available in the data repository for this book.

4.2.1 Predictions

A **hypothesis** is an educated guess about a **mechanism**. The mechanism should be relatively general and could apply to a variety of species or ecosystems. In contrast, a *prediction* is a

specific **measurable result** that will occur in your system *if the hypothesis is correct*.¹

If the hypothesis of Smith-Martin and her colleagues is correct, what should happen in their experiment? What does the hypothesis will predict should happen?

Remember, they set up an outdoor experiment using lianas and trees. The factors that they controlled included,

- the growth habit (lianas and trees),
- species of the lianas and trees, and
- whether they watered to plants or not (all plants were outside and received ambient rainfall).

Of course, they had replicate plants in every condition.

In each plot and for each plant, they measured,

- pre-dawn water potential,
- midday water potential,
- difference between pre-dawn and midday water potential
- photosynthetic rate

They collected these observations in five different months:

- during the dry season (March and April),
- in the transition period between dry wet seasons (May), and
- during the wet season (September and October).

4.3 Predictions

First, describe the hypothesis (i.e. how lianas vs. tress cope with dry periods) in your own words.

Second, state two different predictions about how the factors they controlled would influence the responses they measured. Each prediction should

1. be consistent with the hypothesis, and
2. show the *differences* between lianas and trees.

¹Note these explanations for hypothesis and prediction may differ from those often described in statistics classes.

Remember, they were trying to figure out why lianas are so successful in highly seasonal environments. Explain the logic of each prediction.

Third, draw with `pen(cil)` and paper graphs illustrating your predictions. Take pictures of these graphs so that you can include them in your deliverables. The goal is to place side-by-side the independent variables that should differ - perhaps lianas and trees side-by-side, in a variety of situations.

Your job is to summarize the data in a way to evaluate and communicate the hypotheses. Think about how you can represent the prediction in a graph. *Each graph should use variables that you think will show lianas outperforming trees under the relevant conditions.*

To evaluate a hypothesis and display it visually, we generally put a cause or independent variable on the horizontal x -axis and the response or dependent variable on the y -axis. For example, in the graphs above (Figure 4.2), Schnitzer evaluated and then revealed to us the relations between deterministic causes on the x -axes and the biological responses on the y -axes.

4.4 Methods

4.4.1 Getting organized

First, we need to confirm that we are in the `Rwork` working directory, and import data. In RStudio, you should open your `Rwork` project.

Check the working directory.

```
getwd()
```

Your working directory should be `Rwork`.

Load packages you need.

```
library(ggplot2)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

```
## alternatively you could load the entire tidyverse,  
## which includes ggplot2 and dplyr  
# library(tidyverse)
```

4.4.2 Import the data

1. Go to our [class data repository](#), and download `SmithMartin.csv`.
2. Make sure the file remains in the same simple format, “comma separated values” (.csv).
3. Put `SmithMartin.csv` in your working directory, `Rwork`.
4. Continue to make sure the file remains in the same simple format, “comma separated values” (.csv).

The first five lines contain metadata, so we skip those when we load the data.

```
liana <- read.csv("SmithMartin.csv", # the CSV file name  
                 skip=5 # skip the 5 lines of metadata  
                 )  
## Check the names of variables in the data frame  
## so you know what you have  
names(liana)
```

```
[1] "Plot"           "Tag"            "Season"  
[4] "Month"          "Treatment"      "Habit"  
[7] "Species"        "PredawnMPa"     "MiddayMPa"  
[10] "MiddayMinusPredawnMPa" "Photo.umolCO2"
```

See for yourself the range in the numeric variables in the data set.

```
NumericVars <- liana[, 8:11]  
summary(NumericVars)
```

PredawnMPa	MiddayMPa	MiddayMinusPredawnMPa	Photo.umolCO2
Min. : -3.5500	Min. : -4.942	Min. : -3.3510	Min. : 0.0483
1st Qu.: -0.6990	1st Qu.: -1.918	1st Qu.: -1.3690	1st Qu.: 4.9650
Median : -0.3940	Median : -1.472	Median : -1.0000	Median : 8.9600
Mean : -0.5411	Mean : -1.578	Mean : -1.0365	Mean : 8.9573

3rd Qu.: -0.2060	3rd Qu.: -1.105	3rd Qu.: -0.6438	3rd Qu.: 12.6500
Max. : -0.0200	Max. : -0.197	Max. : 0.0420	Max. : 21.6000
NA's : 11	NA's : 11	NA's : 11	NA's : 912

See for yourself the species of lianas and trees in the data set. This `table()` will count the number of rows associated with each species, whether it is a tree or a liana.

```
# use 'with' to name a data frame
with( liana,
      # do something with variables in that data frame
      table(Species, Habit)
      # close the parenthesis
    )
```

Species	Habit	
	liana	tree
Callichlamyslatifolia	83	0
Crescentiacujete	0	78
Davillakunthii	118	0
Dipteryxoleifera	0	89
Doliocarpusmajor	114	0
Hieronymaalchorneoides	1	81
Huracrepitans	0	65
Machaeriummilleflorum	116	0
Maripapanamensis	105	0
Paulliniapinnata	98	0
Swieteniamacrophylla	0	97
Terminaliaamazonia	0	118

Unless we tell it otherwise, R tends to sort words alphabetically. Months (January, February,...) have an order that is not alphabetical. Here, we make the variable `Month` a little more useful to us by turning it into a *factor* so that we can tell R how to order months.

```
liana <- liana %>% # Start with your data frame
  ## and make the change we want to the Month variable
  mutate(
    Month = factor(Month, levels=c("Mar", "Apr", "May", "Sep", "Oct"))
  )

##...and show the unique values in Month, and also
## show us the sequential order
unique(liana$Month)
```

```
[1] Apr Mar May Oct Sep
Levels: Mar Apr May Sep Oct
```

4.5 Graphics

In this section, we continue to develop our expertise and comfort in making graphs of predictions and also in using the grammar of the `ggplot2` package (Wilkinson 2005; Wickham 2016). `ggplot()` graphics are designed for

- maximum communication: the goal of data display is communication, to ourselves and others,
- transparency of coding: the code should be easy to read, so that the analyst and others can see what has been done, and
- ease of use: the analyst and do a lot of different things relatively easily.

4.5.1 Anatomy of `ggplot()`

In making a graph, we specify,

1. the main function: `ggplot()`
2. the data frame: `data=`
3. the variables for each axis and any conditioning (these are named, oddly, “aesthetics”): `aes(...)`
4. x and y variables: `aes(x=, y=)`
5. conditioning variables, which allow us to show differences between levels of an independent variable : `aes(x=, y=, color=, linetype=,...)`
6. the type of geometric display (e.g., box-and-whisker plot, scatterplot, columns, histogram, line graph, etc.): `geom_boxplot`, `geom_point()`, `geom_line()`, etc.
7. axis labels and other annotations: `labs(x=, y=,...)`
8. facets (how we cut the data up to show different sides of the data): `facet_wrap( ~ groups)` or `facet_grid(row_groups ~ column_groups)`.

So, let’s try some of this.

4.5.2 Examples of `ggplot()`

Here we go.

A simple plot.

```
ggplot(data=liana, # we specify the data frame
  aes(x = Month, # x-axis
    y= PredawnMPa # y-axis
  ) ) + # close parentheses and add a plus sign to add box and whisker plot
  geom_boxplot()
```

Warning: Removed 11 rows containing non-finite outside the scale range (``stat_boxplot()``).

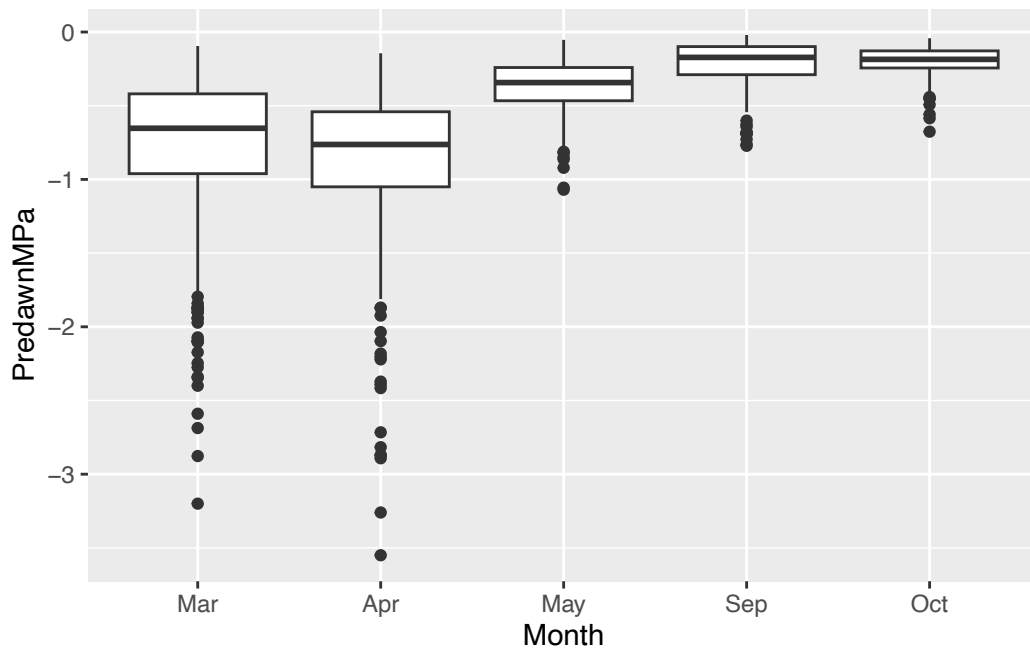


Figure 4.4: Example of a simple box and whisker plot in `ggplot()`. Note the correspondance between the code and the figure. The data frame is 'liana'. The x-axis is defined by `x = Month`. The y-axis (predawn water potential, in megapascals) is defined by `y=PredawnMPa`.

Coding: Note the correspondence between the graph and the elements of the code that made the graph.

Ecology: Note the difference in predawn water potential in March and April, compared to September and Oct. What is water potential, and how does it change from the soil, to the root, stem, and leaf, and finally to the air? Remember that water moves from areas of higher moisture to areas of lower moisture, that is, *down gradients of water potential* from higher values to lower value, just like a ball rolling downhill. In which months is the soil driest?

Now let's try a more complex figure, in which we add two conditioning variables, and more description label on the *y*-axis. We color-code the first conditioning variable is the *treatment*, in which plants are watered or not. The second conditioning variable is *season* (dry, wet or transition), and we make separate *facets* or panels for that.

```
## specify the data and most of the variables
ggplot(data=liana, # we specify the data frame
  aes(x = Habit, # x-axis
    y= MiddayMPa, # y-axis
    color = Treatment # conditioning variable. Examine the output to figure out what t
  ) ) + # close parentheses and add a plus sign to add box and whisker plot
  geom_boxplot() + # and add "faceting" to break it up by season.
  facet_wrap(~Season) + # the season
  labs(y="Midday water potential (MPa)")
```

Warning: Removed 11 rows containing non-finite outside the scale range (``stat_boxplot()``).

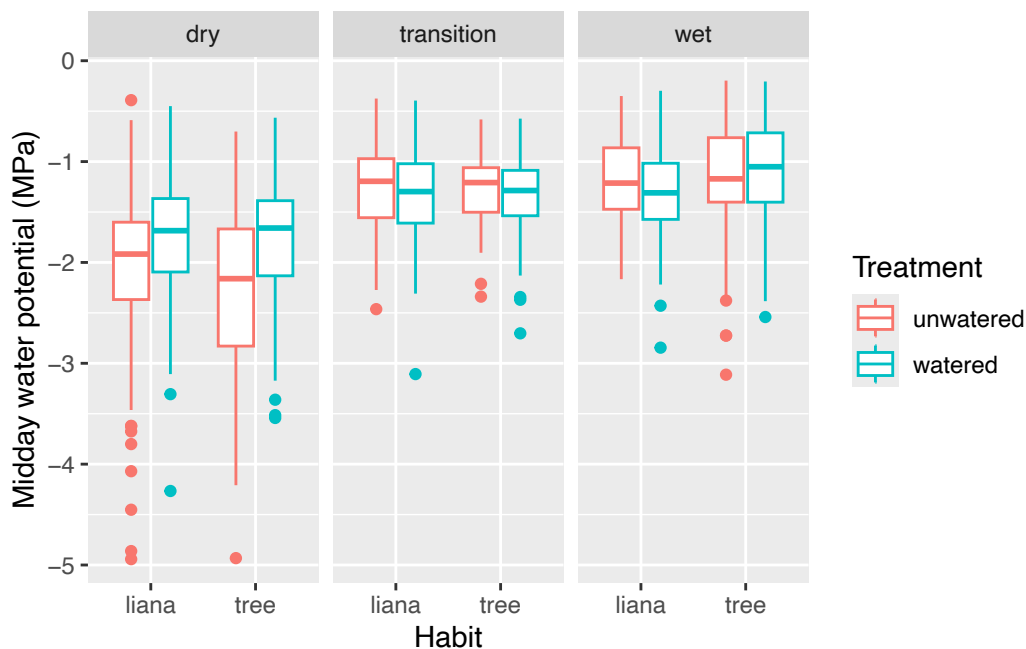
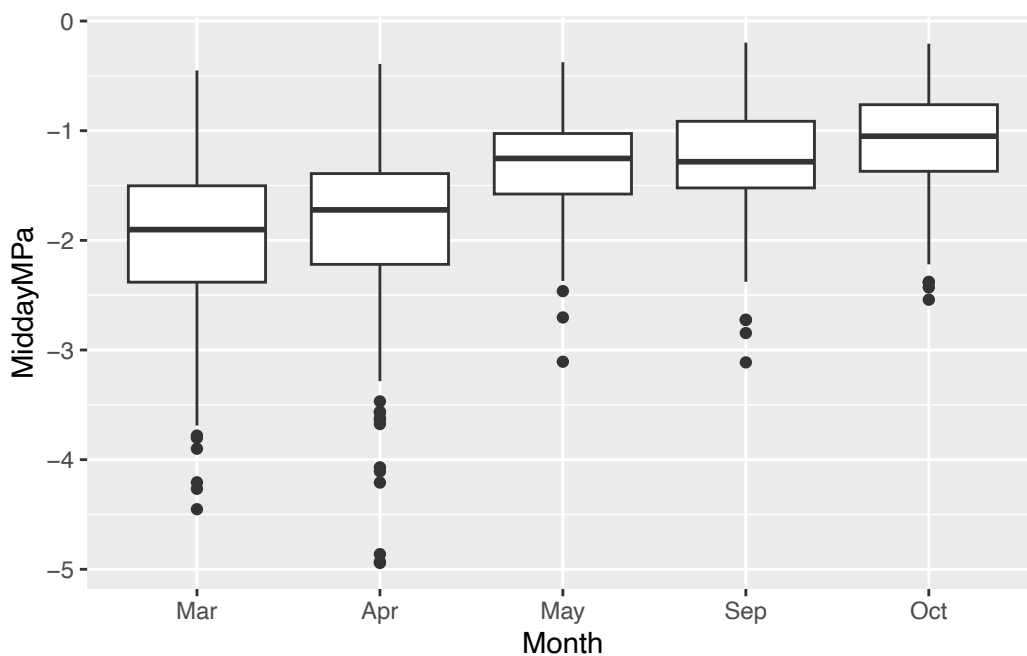


Figure 4.5: Example of a box and whisker plot in `ggplot()`. Note the correspondance between the code and the figure. The x-axis is defined by `x = Habit`. The y-axis (midday water potential, in megapascals) is defined by `y=MiddayMPa`. The data are further split by experimental treatment (watered vs. not watered), and that is defined by `color = Treatment`.

And remember that you can save a graph to your working directory using `ggsave()` right after you make your graph, like this.

```
ggplot(data=liana, # we specify the data frame
  aes(x = Month, # x-axis
    y= MiddayMPa, # y-axis
  ) ) + # close parentheses and add a plus sign to add box and whisker plot
  geom_boxplot()
```

Warning: Removed 11 rows containing non-finite outside the scale range (``stat_boxplot()``).



```
ggsave("myMiddayPlot.png", width=4, height=4)
```

Warning: Removed 11 rows containing non-finite outside the scale range (``stat_boxplot()``).

The dimensions (in inches) of your *saved* graph dictate how big the labels and points look. If you make a small graph (e.g., width=3, height=3), the labels will look big by comparison. If you make a large graph (e.g., width=10, height=10), the labels will look small by comparison. Try it, and see for yourself.

4.6 Deliverables

Turn in two documents, one document has your ideas, graphs, and answers, and the other document is your R script.

A. The first document will contain:

1. A restatement, in your own words, of the hypothesis of Smith-Martin and her colleagues.
2. Two predictions of the hypothesis that you can evaluate using the data we have (see above). You should take pictures of these hand-drawn images, and put them into this document.
3. Two graphs done in R that use data to evaluate the hypothesis of Smith-Martin and her colleagues. The variables in the graphs must match the variables in your predictions.
4. For each graph, identify the independent variable(s) and the dependent variable(s).
5. For each graph, explain what the graph shows and what it tells us about the hypothesis. We can't draw firm conclusions, so use terms such as "suggests", "is consistent with", "seems to show", or others, which help avoid expressing unwarranted certainty. Even with careful analysis, we always have to be cautious.

B. Your second document is your R script. The R script should contain *everything a person would need* aside from the data itself to make the figures that you turn in, as well as identifying information. For instance, this should include:

- Your name, the course, the chapter or assignment, and the date, all as comments with leading hastags (#s).
- Loading any packages you need, such as `ggplot2`, or others.
- code to load (or "read") any data you need, and comments that explain what the code does,
- code to organize the data if you need to, and comments that explain what the code does,
- code to create, and to save any graphs you make, and comments that explain what the code does.

5 Behavior: Extra-pair copulation in a monogamous species



Figure 5.1: Voles are small hamster-like rodents that live in grasslands and fields.

5.1 Goals

In this assignment, you will

- Develop and use your understanding of evolution, behavior, and prairie voles to make predictions about determinants of extra-pair copulation in a monogamous species.
- Enhance your quantitative reasoning and data analytic skills.

5.2 Preparatory Steps

Download Keane et al. (2024), which is available in [our PDF archive](#). Read the introduction to help you understand the context of this chapter.

5.3 Background: What causes extra pair copulation in prairie voles?

Monogamy, the mating system in which individuals pair-bond to raise offspring, is relatively rare. We don't know what causes this rarity, and nor do we know why it persists where it does occur. Prairie voles (*Microtus ochrogaster*), like humans, are famously monogamous, and behavioral ecologists have studied them for that reason. And, just like humans, voles also engage in extra-pair copulations that result in offspring, and scientists have been trying to understand the causes and consequences of those extra-pair copulations (Solomon and Keane 2018).

Monogamy is a mating system (*sexual monogamy*), but it is also one of many types of social system (*social monogamy*). Most species are entirely *solitary* (think grasshoppers, frogs, flies), but some species engage in persistent social behavior, or exhibit *sociality* (Nowak, Tarnita, and Wilson 2010; Nordell and Valone 2014; Kappeler 2019). For example, all species of ants and many species of bees live in large colonies with several castes of individuals (*eusociality*). Many species live in family social units or have primitive forms of eusociality (Nordell and Valone 2014). Social monogamy is the particular form of sociality in which a female-male pair lives together, often resulting in more successful rearing of offspring.

So why would socially monogamous individuals engage in extra-pair copulation?

One reason that individuals engage in extra-pair copulations is in order to avoid inbreeding depression. Breeding that results from matings between highly related individuals typically causes a decline in vigor and health of individual offspring and a reduction in fitness for those phenotypes. Inbreeding depression is the reduction in fitness due to breeding among relatives.

Fitness is the quantitative representation of natural and sexual selection within evolutionary biology. It can be defined either with respect to a genotype or to a phenotype in a given environment. In either case, it describes individual reproductive success and is equal to the average contribution to the gene pool of the next generation that is made by individuals of the specified genotype or phenotype. The fitness of a genotype is manifested through its phenotype, which is also affected by the developmental environment. The fitness of a given phenotype can also be different in different selective environments. (Quoted from [Wikipedia](#))

5.3.1 Tinbergen's four questions

This section is from an [online course at Oxford University](#)

In 1963, Niko Tinbergen published a seminal paper entitled *On the aims and methods of ethology*, which laid the foundation for how to conduct research in the fledging field of animal behaviour. The lasting contribution of the paper is that in it Tinbergen formulated four different, albeit somewhat interlinked, approaches to the study of animal behaviour, or four different types of questions we can ask about an observed behaviour.

The four questions are:

1. Function (or adaption)

Why is the animal performing the behaviour? In which way does the behaviour increase the animal's fitness (i.e. its survival and reproduction)? Examples are plentiful and include, among many others, nurturing of young to increase their chance of survival, migration to warmer (and more food rich) habitats, escaping or avoiding attention from predators etc.

2. Evolution (or phylogeny)

How did the behaviour evolve? How has natural selection modified the behaviour over evolutionary time? This is typically addressed by the comparative approach, where the behaviour in question is compared among closely related species. Examples include how flight in birds may have evolved from gliding in dinosaurs or how the vertebrate and cephalopod eyes have evolved by convergent evolution, with the former having a blind spot, while the latter does not.

3. Causation (or mechanism)

What causes the behaviour to be performed? Which stimuli elicit or what physiological mechanisms cause the behaviour? Examples include the role of pheromones and hormones, such as increasing testosterone levels (caused by increasing day length) causing male display behaviour in many species of birds, moving shadows causing ragworms to withdraw into their burrows or contrast on beaks causing herring gull chicks to peck.

4. Development (or ontogeny)

How has the behaviour developed during the lifetime of the individual? In what way has it been influenced by experience and learning? Examples include how courtship behaviour improves with age in many birds and how predators learn to avoid toxic or dangerous prey with experience.

The four types of questions operate at two different levels. Questions 1 and 2 give ultimate or evolutionary explanations. These are answers that take a longer perspective and try to explain *why* the behaviour has evolved. On the other hand, questions 3 and 4 give proximate

explanations. These are answers that look into the immediate mechanical reasons for *how* a behaviour is expressed. To obtain a full understanding of the costs, benefits and constraints that have shaped a given behaviour both ultimate and proximate answers must be obtained.

5.3.2 Determinants of extra-pair copulation in voles

So, what causes extra-pair copulation? What influences the probability that an individual engages in extra-pair copulation? We can think about these questions in any of the four contexts that Tinbergen's questions provide.

Here are some factors that may help determine to the probability of extra-pair copulation:

- the relatedness of the extra-pair partner, and avoidance of inbreeding.
- the relatedness of the nest- vs. extra-pair partner.
- sex (perhaps females are more or less likely than males to seek opportunities)
- number in nest, if crowding is a proximate cue to mate with a nest partner or extra-pair individual.

It is not always clear how animals and plants identify individuals that are closely related or not related, but under many circumstances, they somehow seem to mate with individuals that are less related than by chance alone.

5.4 Data from the Solomon/Keane lab

Dr. Nancy Solomon and Dr. Brain Keane have been studying prairie voles for decades, and here we investigate our questions using one of their data sets.

The data file `epr_209.csv` contains observations on voles related to those we read about in Keane et al. (2024). These data are from sampling of voles from the Indiana site, collected over three years (2006-2008). These data are available in [our data repository](#)

Each row in the data set is a record of a succesful copulation that resulted in at least one offspring. Included in each row we have the offlowing columns:

- id - identification number of the individual
- sex - biological sex of the individual
- year - calendar year of data collection
- nest - the identity of the home nest of the individual
- size - number of individuals in the nest
- r - relatedness to the partner.
- pairtype - nest partner or extra-pair copulation.

	id	sex	year	nest	size	r	pairtype
1	101	male	2006	19/20/21	7	0.3776	nest
2	101	male	2006	19/20/21	7	-0.1830	nest
3	1021	female	2007		29	3 0.3208	nest
4	1021	female	2006		32	2 -0.1093	extra

Warning in rm(d209): object 'd209' not found

Relatedness to a partner varies between -1 and +1. A value near zero indicates no more related than expected by chance; 1 is self-identity (e.g., clone) and negative values indicate even less related than we expect due to chance.

5.5 Your objectives

Ask yourself, “what the heck are these voles doing?” What is might be a cause of extra-pair copulations?

You consider what you learned in your readings, and come up with a question or hypothesis about what conditions to might lead to more extra-pair copulations. In particular, you will

1. Hypothesis: Pose a hypothesis that you can address *using these data*. Your hypothesis should reflect a mechanism of what you think is going on. Rely on your understanding of evolution, behavior, and prairie voles to justify your the mechanism and hypothesis.
2. Prediction: Make a prediction that is based on your hypothesis. Draw *on paper* a mock-up of the graph that uses the sort of data we have access to, and is consistent with the mechanism you propose.
3. Data: Write R code to create a graph that mimicks your prediction, but uses the actual data you download.
4. Evaluation: Figure out what the graph tells you about the answer to your question. In what ways are your results consistent with your prediction or not? Rely on your understanding of evolution, behavior, and prairie voles to interpret your result.

Put all of the above (hypothesis, hand-drawn prediction, R figure of data, data-based evaluation) in a document, and turn it in online. See **Deliverables** at the end of this chapter.

5.5.1 Write your own script

To create the graph, you will write your own script. Start as usual, by finding out what your current working directory. Make sure it is **Rwork**. The read in your data and look at the variables. Then create the graph you want.

First we check the current working directory.

```
getwd()
```

This should show a path to `Rwork`. If not, use the “Session” menu to set your working directory. If you don’t have an `Rwork` folder, see Chapter 1.

Next we load some packages that we will use.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v ggplot2     4.0.2      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr       1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Next, we get the data.

1. Go to our [class data repository](#), and download `epr.csv`.
2. Make sure the file remains in the same simple format, “comma separated values” (.csv).
3. Put `epr.csv` in your working directory, `Rwork`.
4. Continue to make sure the file remains in the same simple format, “comma separated values” (.csv).

Now import the data, and display its *structure*.

```
d209 <- read.csv("epr_209.csv")
str(d209)
```

```
'data.frame':  250 obs. of  7 variables:
 $ id      : chr  "101" "101" "1021" "1021" ...
 $ sex     : chr  "male" "male" "female" "female" ...
 $ year    : int   2006 2006 2007 2006 2006 2007 2007 2006 2006 2006 ...
 $ nest    : chr  "19/20/21" "19/20/21" "29" "32" ...
 $ size    : int    7  7  3  2  2  2  2  7  7  2 ...
 $ r       : num   0.378 -0.183 0.321 -0.109 -0.032 ...
 $ pairtype: chr   "nest" "nest" "nest" "extra" ...
```

Use everything you have learned thus far in previous chapters, and the tutorial below to make the graph you want. After you make the graph you want, you can use `ggsave("myVoles.png")` to save the graph to your working directory.

5.5.2 Recoding “pairtype”

We can estimate the probability of something happening or not happening based on a series of successes and failures, like a coin coming up heads. A coin either comes up heads or it doesn't - success or failure. We can estimate the probability that it comes up heads ($P = 0.5$), even though we have only heads or tails, successes or failures.

We need to create a variable that represents either an extra-pair copulation or a nest-pair copulation. We will code extra-pair copulation as 1 (success) and nest-pair copulation (failure).

```
d209 <- d209 %>%
  mutate(
    epc = ifelse(pairtype=="extra", 1, 0)
  )
str(d209)
```

```
'data.frame': 250 obs. of 8 variables:
 $ id      : chr  "101" "101" "1021" "1021" ...
 $ sex     : chr  "male" "male" "female" "female" ...
 $ year    : int   2006 2006 2007 2006 2006 2007 2007 2006 2006 2006 ...
 $ nest    : chr  "19/20/21" "19/20/21" "29" "32" ...
 $ size    : int   7 7 3 2 2 2 2 7 7 2 ...
 $ r       : num   0.378 -0.183 0.321 -0.109 -0.032 ...
 $ pairtype: chr  "nest" "nest" "nest" "extra" ...
 $ epc     : num   0 0 0 1 1 1 1 0 0 1 ...
```

Now we see that extra-pair copulation is coded as a number, just 0 or 1.

See below for an example of graphing zeros and ones on the y-axis, and a continuous variable on the x-axis, where we describe the probability that a car model has front wheel drive (y-axis), based its city gas mileage.

5.6 Graphing in ggplot()

Here is some general guidance for using `ggplot()`:

```
ggplot(data= myDataset, aes(x= X-variable, y= Y-variable)) + geom_typeOfPlot()
```

- for a box-and-whisker plot: `geom_boxplot()` This is a more transparent and useful alternative to standard barplot or column graph.
- for a scatterplot `geom_point()`
- for a scatterplot with a fitted straight line `geom_point() + geom_smooth(method="lm")`

First, we load the packages of the tidyverse. If you already loaded this package as part of `tidyverse`, then you have already done this step.

```
library(tidyverse)
```

Here we use a data set (`mpg`) that comes with R to help demonstrate examples. This data set, `mpg` includes fuel economy data from 1999 to 2008 for 38 popular models of cars.

```
glimpse(mpg)
```

```
Rows: 234
Columns: 11
$ manufacturer <chr> "audi", "audi", "audi", "audi", "audi", "audi", "audi", "~
$ model        <chr> "a4", "a4", "a4", "a4", "a4", "a4", "a4", "a4 quattro", "~
$ displ       <dbl> 1.8, 1.8, 2.0, 2.0, 2.8, 2.8, 3.1, 1.8, 1.8, 2.0, 2.0, 2.~
$ year        <int> 1999, 1999, 2008, 2008, 1999, 1999, 2008, 1999, 1999, 200~
$ cyl         <int> 4, 4, 4, 4, 6, 6, 6, 4, 4, 4, 4, 6, 6, 6, 6, 6, 8, 8, ~
$ trans       <chr> "auto(l5)", "manual(m5)", "manual(m6)", "auto(av)", "auto~
$ drv         <chr> "f", "f", "f", "f", "f", "f", "f", "4", "4", "4", "4", "4~
$ cty        <int> 18, 21, 20, 21, 16, 18, 18, 18, 16, 20, 19, 15, 17, 17, 1~
$ hwy        <int> 29, 29, 31, 30, 26, 26, 27, 26, 25, 28, 27, 25, 25, 25, 2~
$ fl         <chr> "p", "p", "p", "p", "p", "p", "p", "p", "p", "p", "p", "p~
$ class      <chr> "compact", "compact", "compact", "compact", "compact", "c~
```

For a box-and-whisker plot of highway mileage (`hwy`) vs. car type (`class`), with explicit labels.

```
ggplot(mpg, aes(x = class, y = hwy)) + geom_boxplot() +
  labs(x="type of car", y="highway mileage")
```

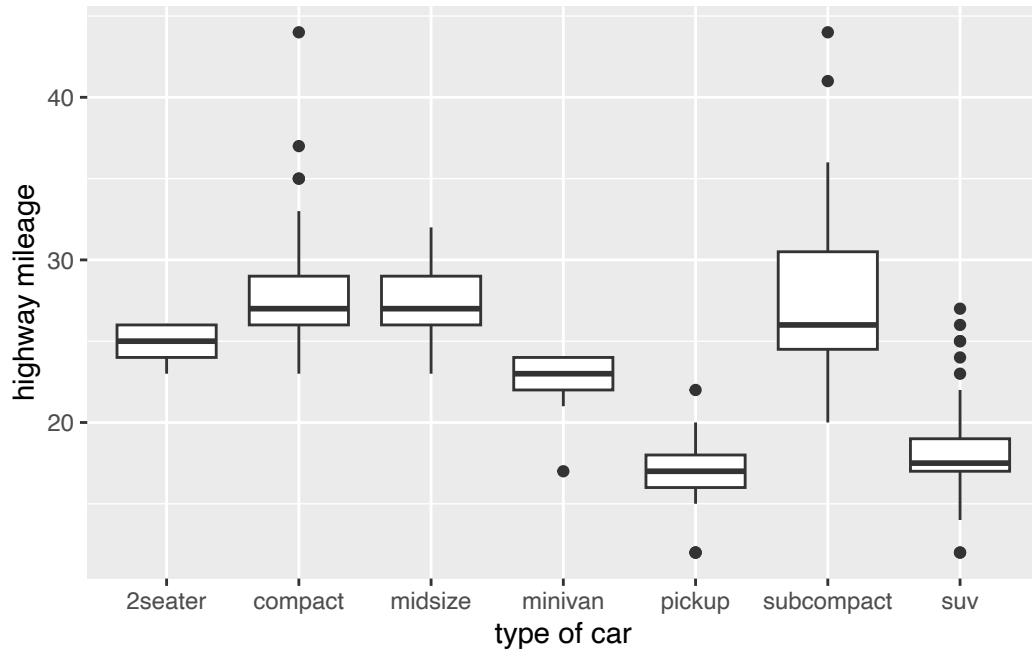


Figure 5.2: Box-and-whisker plot of highway mileage (hwy) vs. car type (class), with explicit labels. A boxplot shows the distribution of data, by groups.

5.6.0.1 Scatterplots

```
ggplot(mpg, aes(x=cty, y=hwy)) + geom_point() +  
  labs(x="city mileage", y="highway mileage")
```

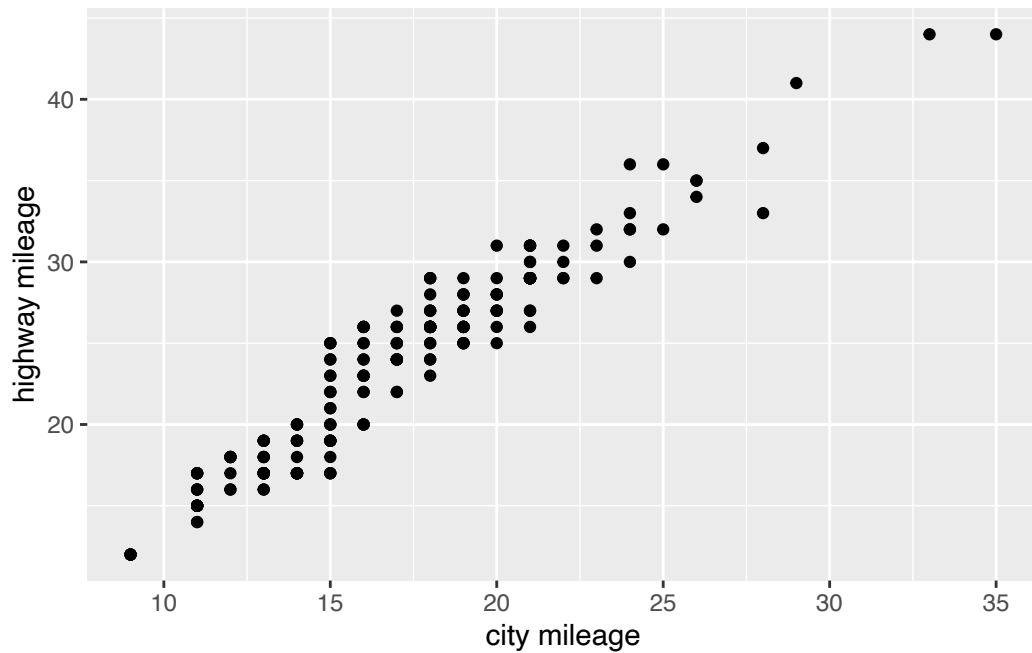


Figure 5.3: Scatterplot of highway mileage vs. city mileage (miles per gal).

```
ggplot(mpg, aes(x=cty, y=hwy)) + geom_point() + geom_smooth(method="lm")
```

``geom_smooth()`` using formula = 'y ~ x'

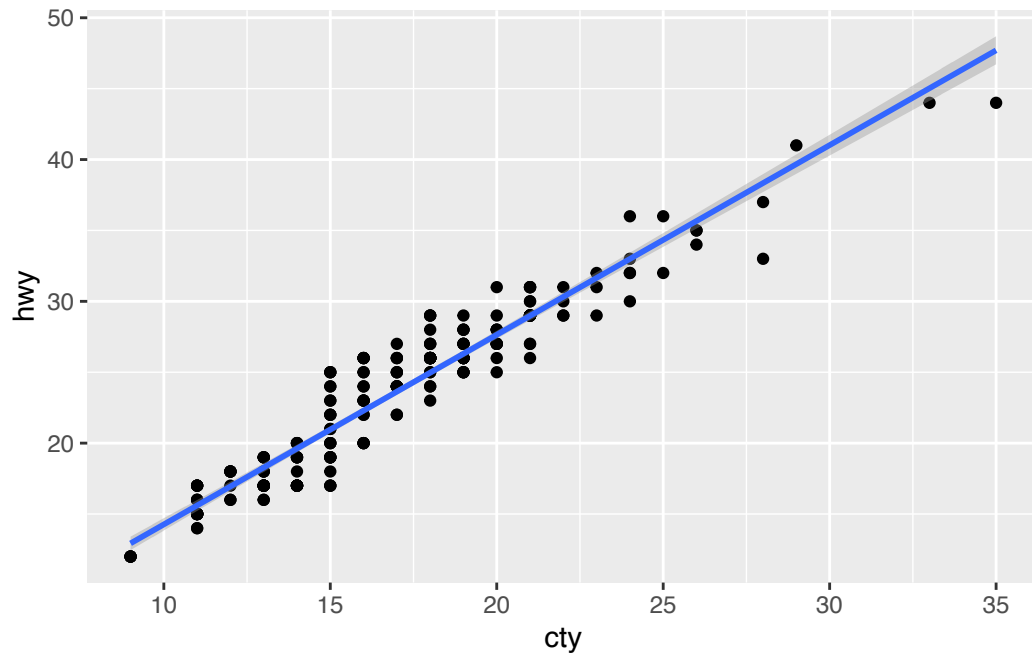


Figure 5.4: Scatterplot with a fitted straight line, fitting a linear model.

```
## for a scatterplot with a fitted curved line  
ggplot(mpg, aes(x=cty, y=hwy)) + geom_point() + geom_smooth()
```

``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

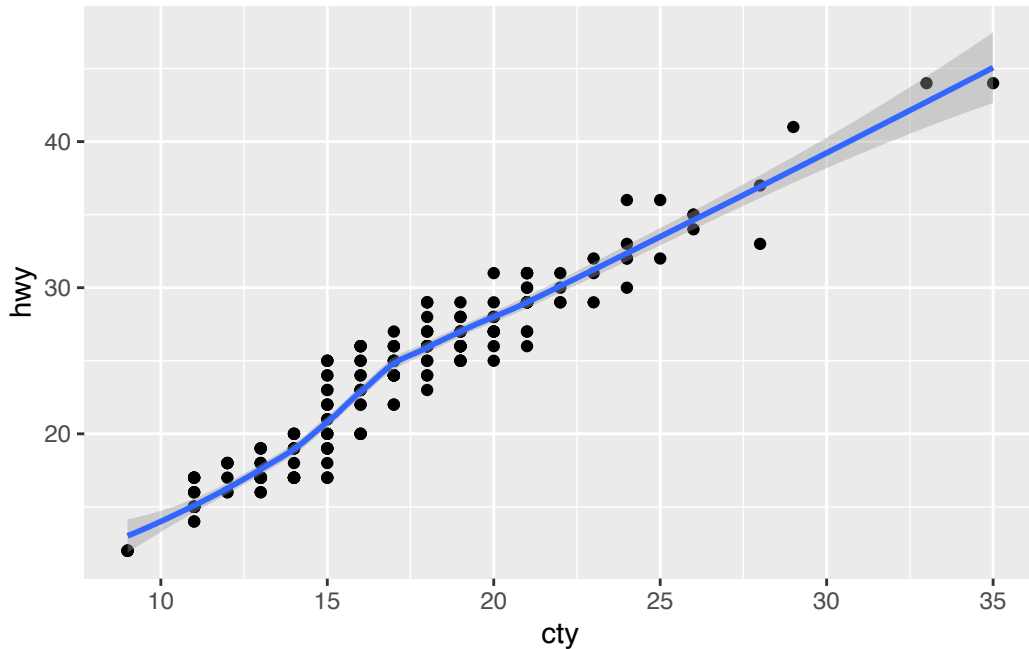


Figure 5.5: Scatterplot with a flexibly fitted line, using local polynomial regression fitting.

5.6.0.2 Categorical responses

Often we would like to predict whether individuals fall into *one category or another*, such as coin flips of heads vs. tails, success vs. failure, sick vs. healthy, or engaging in extra-pair copulation vs. remaining monogamous. These kind of dichotomous data are called *binomial* data, and we think of these data as zeros and ones (0, 1), perhaps zero for heads and one of tails.

When we model binomial data (0,1), we put these 0s and 1s on the y-axis. Here is a worked example using `mpg` data.

First, create a dichotomous variable (0, 1) indicating whether a car model is front wheel drive or other (rear- or 4-wheel drive). A plot like this puts *probability* on the y-axis. In this case it is the probability that a car with a given mileage is a front-wheel drive car.

Convert a factor or character variable to zeros and ones. Here we convert the category of drive train (rear vs. front wheel drive) into ones for front wheel drive, and zeros for rear wheel drive. We use an “if-else” statement to do that.

```
mpg2 <- mpg %>%
  mutate(
    probb.front.wheel.drive = ifelse(drv=="f", 1, 0)
  )
```

Next we plot the zeros and ones on the y-axis against city mileage on the x-axis. We also add a particular kind of fitted line that shows us the expected probability that a car with a particular city mileage is front wheel drive. This curve is the result of a *logistic regression model* that we use for binomial data. The graph tells us that

```
ggplot(mpg2, aes(cty, prob.front.wheel.drive)) +  
  geom_point() +  
  geom_smooth(method="glm",  
              method.args = list(family = "binomial"))
```

`geom\_smooth()` using formula = 'y ~ x'

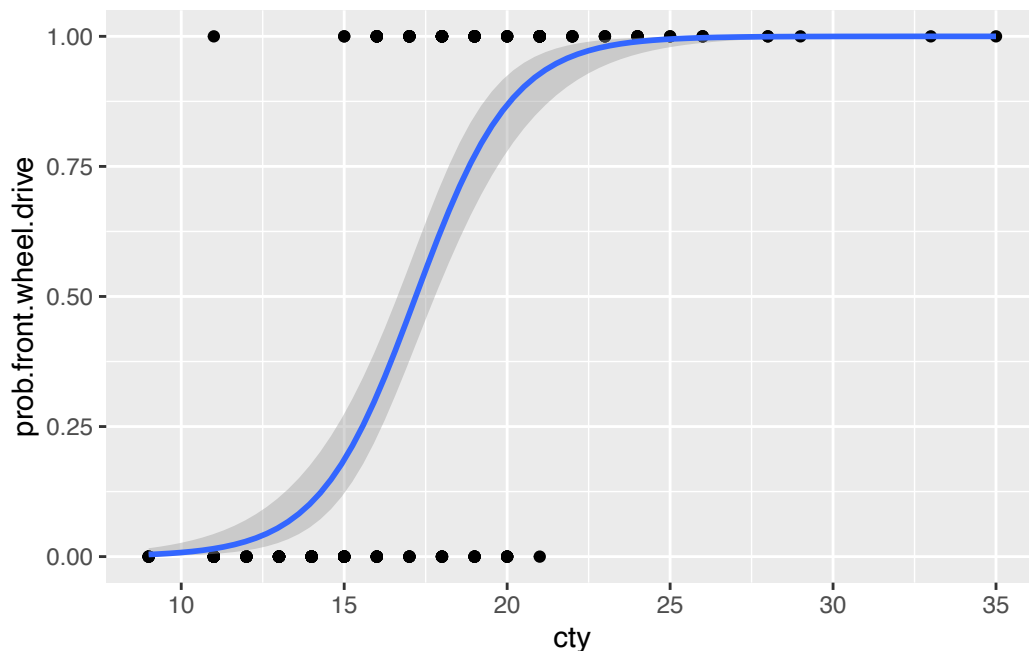


Figure 5.6: Higher mileage cars tend to have front-wheel drive. Graph shows the probability that a car with a particular city mileage is front wheel drive. The curve is the result of a type of generalized linear model (glm) called *logistic regression* which assumes binomial data.

5.7 Deliverables

Create one document (presentation slide or other document) that,

- places the hand-drawn graph of your predictions side by side with the graph of the observed data,
- includes a succinct *explanation* of
 - i. basis of your original prediction; explain why voles would behave the way you predicted they would. Do not rely on your intuition. Instead, use your knowledge of evolutionary biology, behavior, and voles in your explanation, and
 - ii. your observed result.
- includes a *headline*, that is, a title that provides your result or takeaway message, such as “Living in small groups enhances monogamy”.

Turn this in to the assignment on Canvas. (*Previous versions asked to submit this to a Google Slide deck, but we will no longer do that.*)

6 Density-independent Population Growth: Humans on the rise

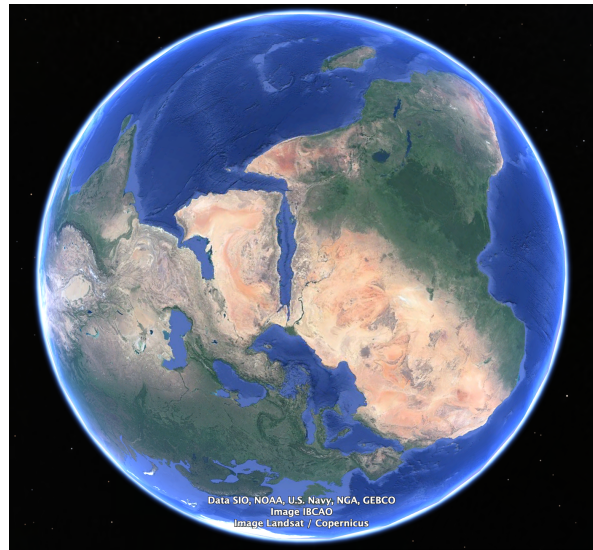


Figure 6.1: Earth has now entered the Anthropocene. Geologists are debating precisely when that happened.

Reproduction helps define what we call “life”, and leads to exponential growth. This chapter introduces this concept and how we quantify it.

6.1 Goals

In this chapter, you will

- learn the two fundamental principles of population growth,
- do a little math: practice calculating rates of population growth,
- do a little programming: use a “for-loop” to do lots of work very easily,
- determine how the rate of human population growth has changed over time, and
- start to develop a greater understanding and intuition for exponential rates.

In this chapter, we'll use these techniques and data from *Our World in Data* to describe mathematically the growth rate of the world's human population at different points in time over the last 10,000 years.

6.2 Preparatory Steps

- Read Cohen (1995) in Perusall through our Canvas site. It is also available in our class archive: [Cohen \(1995\)](#)
- 1. Go to our [class data repository](#), and download `HumanPopData-2023-0101.csv`. Make sure the file remains in the same simple format, “comma separated values” (.csv).
- If you are curious, check Deliverables, at the end.

6.3 Background

Given sufficient resources, life is autocatalytic – the products of reproduction (offspring) amplify the rate of reproduction, leading to *exponential growth*.

As ecologists, we can state this with the following propositions (Hastings 2011):

1. *A population grows exponentially in the absence of other forces.*
2. *There are forces that can prevent a population from growing exponentially.*

This deceptively simple pair of propositions provides a basis for an enormous amount of ecology. They are analogous to Newton's First Law of Motion:

Every body perseveres in its state of being at rest or of moving uniformly straight forward, except insofar as it is compelled to change its state by forces impressed. (*The Principia: Mathematical Principles of Natural Philosophy* (1687), 3rd edition (1726), trans. I. Bernard Cohen and Anne Whitman (1999), Axioms, or Laws of Motion, Law 1, 416.)

Now we would say that *a body remains at rest, or in motion at a constant speed in a straight line, unless acted upon by a force.*

The first proposition of population growth can emerge as the consequence of the following assumptions:

- a. all individuals in a population are identical.
- b. there is no migration in or out of the population.
- c. i. the number of offspring per individual (or the per capita birth and death rates) are constant through time, and (ii) independent of the number of individuals in the population.

Deviations from these assumptions lead to proposition 2, and most of the interesting parts of single species population dynamics (Hastings 2011).

However, although propositions 1 and 2 can follow from assumptions a-c, these assumptions are not necessary. For instance, in some cases, individuals don't need to be identical, and we deal with a big exception in the next chapter where we introduce structured population growth. Also, migration is admissible, provided immigration = emigration and that it doesn't alter the rate of change. Nonetheless, other deviations from a and b can have very important consequences for single species population dynamics. We will get to those in later chapters.

Even when populations don't grow exponentially, we often describe population growth using the same mathematical approach. We do that for at least a few of reasons:

- approximations of more complicated dynamics often simplify to exponential growth under reasonable assumptions,
- over short periods of time, deviations to exponential growth still result in patterns that are very similar to exponential growth, and
- we typically don't know precisely what mechanisms limit exponential growth or by how much, and therefore exponential growth is as good a mathematical description as any.

6.3.1 Population growth of *Homo sapiens*: a case study

The human population shares many features with other populations. Like other species, humans are distributed unequally around our available habitats, we consume resources and produce waste products, and the size of our population changes over time.

Humans, as we would recognize ourselves, have been around for ~300,000 years. Since that time a lot has happened to us, and we have done a lot to alter the planet. About 10,000 years ago, we started to use agriculture to raise a lot of our food, in the *agricultural revolution*. Oddly, this seems to have contributed simultaneously to both a decline in life span due to poor health, and yet also an increase in population growth rate and a corresponding increase in population size. The state of the human population is dynamic and undergoing transitions that are in different stages in different places around the world (Lee 2011).

Here we examine changes in human population size from 10,000 y ago to 2021. The data are estimates, and we have annual estimates with increasing greater frequency, up until 2021.

We will calculate the density-independent per capita growth rate for pairs of years in the data set. We'll do this by written out by hand, and also in R, using code to do everything quickly, efficiently and, importantly, reproducibly, so that you—or anyone—could check your estimation procedure.

We often use estimates of λ and r to quantify how rapidly a population is growing (or declining) at various points in time. For a population with overlapping generations and reproducing

continuously, we can estimate the *per capita* exponential growth rate at a particular point in time (r_t) using the following equation:

$$r_t = \frac{\frac{\ln N_{t+h}}{\ln N_t}}{(t+h) - t} \quad (6.1)$$

where N_t is the size of the population at some starting time point t , and N_{t+h} is the size of the population after some time interval h , and $\ln$ is the natural logarithm.

For example, imagine we that in 2020, we had 100 mice and exactly two years later, the population had grown to 150 mice. Here are the values that we would use to calculate r_t :

- $t = 2020$
- $t + h = 2022$
- $h = (t + h) - t = 2022 - 2020 = 2$
- $N_t = 100$
- $N_{t+h} = 150$

We would substitute these values into the above formula (#eq-r) as,

$$\begin{aligned} r_t &= \frac{\ln \frac{150}{100}}{2022 - 2020} \\ &= \frac{\ln 1.5}{2} \\ &= 0.2027 \end{aligned}$$

showing each step of our work. Whenever you need to show how you arrived at an answer (such as in homework, research, or in a job), you should do something like this.

When r_t is positive ($r_t > 0$), this means that the population was growing, and quantifies how *fast* it grew and became larger.

What if we reverse so the we started with $N_t = 150$ and the population declined to $N_{t+h} = 100$?

$$\begin{aligned} r_t &= \frac{\ln \frac{100}{150}}{2021 - 2020} \\ &= \frac{\ln \frac{2}{3}}{2} \\ &= -0.2027 \end{aligned}$$

ending with a negative value ($r_t < 0$), indicating that the population shrank. When $r_t < 0$, that means the population has decreased in size. Sometimes we call that “negative growth”.

What do you think we get if the population didn’t change, so that $N_t = N_{t+h} = 100$?

6.3.2 Projecting population size into the future

If we make certain assumptions, we can make predictions about the future. In order to make predictions about the size of the population next year, or in 5, 10, or 20 years, we can extrapolate the current rate of change into the future. To do that, we can use the instantaneous rate of increase, r , and the amount of time, and a solution from calculus.

To predict the size of a population to year t from some starting time, we rely on the rule that,

$$N_t = N_0 e^{rt} \quad (6.2)$$

or equivalently,

$$N_{t+h} = N_t e^{rh}$$

where the parameter r is the per capita instantaneous rate of increase, and t is the amount of time, or time interval. The letter e is the exponential, or Euler’s constant. The exponential is the base of the natural logarithm. **The important thing here is the time interval in the exponent (t or h) has to be the time interval from the starting value of N to the projected value of N .**

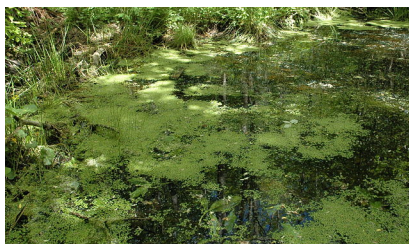


Figure 6.2: Common duckweed (*Lemna minor*) populations can grow fast. Credit: <https://commons.wikimedia.org/w/index.php?curid=1085476>

Imagine we have a population of the small floating plant duckweed (*Lemna minor*), that can double in about two weeks. In that case, $r = 0.35$ new lemna per old lemna per week. Pretend we start with a population of 100 lemna individuals ($N_0 = 100$). How many lemna should we have in three weeks? Show your work (e.g., Figure 6.3).

(Overthinking: Equation 6.2 is derived from the growth equation Equation 6.1 using integral calculus. It is the integral from 0 to t of Equation 6.1.)

$$r = 0.35 ; N_0 = 100 ; \text{time interval} = 3 \text{ week}$$

$$N_t = N_0 e^{rt} = 100 e^{0.35 \cdot 3} = 285.8 \text{ individuals}$$

Figure 6.3: Projecting the size of the duckweed population three weeks into the future.

6.4 Methods

1. Make sure your working directory is `Rwork`.
2. Load the necessary packages.

```
library(tidyverse) # for readr, dplyr and others
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v ggplot2     4.0.2      v tibble     3.3.0
v lubridate   1.9.4      v tidyr      1.3.1
v purrr       1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(patchwork) # to combine ggplot2 figures
```

6.4.1 Load data

Next, we get the data, if you haven't already done so.

1. Go to our [class data repository](#), and download `HumanPopData-2023-01-01.csv`. Make sure the file remains in the same simple format, “comma separated values” (.csv).
2. Put `HumanPopData-2023-01-01.csv` in your working directory, `Rwork`.
3. Open `HumanPopData-2023-01-01.csv` and make note of the first three lines of metadata. When you close the file, do not save any changes, and make sure *not* to save it in a new format, such as “.xlsx” or “.numbers”.
4. Continue to make sure the file remains in the same simple format, “comma separated values” (.csv).

Now import the data into R, and *glimpse* the structure of the data frame.

```
d <- read_csv("HumanPopData-2023-01-01.csv",
              skip=3 # skip three lines of metadata
              )
```

```
Rows: 259 Columns: 2
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
dbl (2): Year, est.N
```

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
glimpse(d)
```

```
Rows: 259
```

```
Columns: 2
```

```
$ Year <dbl> -10000, -9000, -8000, -7000, -6000, -5000, -4000, -3000, -  
2000, ~
```

```
$ est.N <dbl> 4432266, 5616996, 7242892, 9577918, 13201832, 19075771, 28774661~
```

Note that years before the current era (BCE) are negative numbers, while numbers in the current era (CE) are positive.

Let's also graph the data to see if it makes sense.

```
d %>%  
  ggplot(aes(x=Year, y=est.N)) +  
  geom_point() + geom_line() +  
  scale_y_log10() + labs(y="Estimated human population size")
```

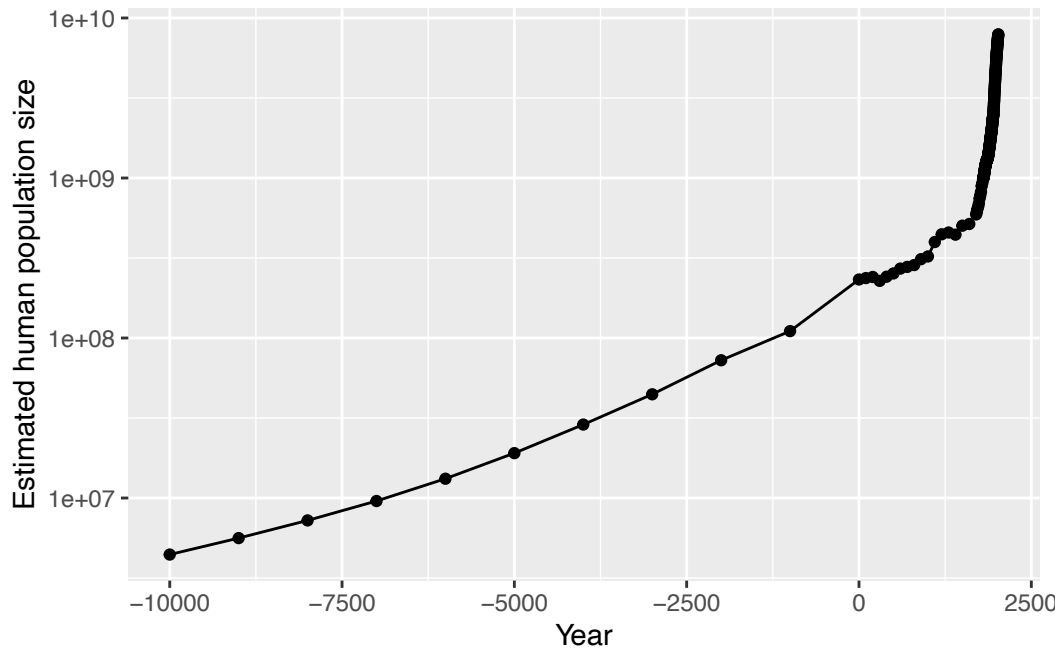


Figure 6.4: The human population has increased over time. Note the common log scale on the y-axis. Note that years before the current era (BCE) are negative numbers, while numbers in the current era (CE) are positive.

6.4.2 Simple math: Calculations by hand

Doing math and arithmetic by hand is like learning music: you have to do it by hand, even when it feels like copying. This helps it become automatic, which leads to understanding.

Calculate by hand the instantaneous per capita growth for these intervals:

- 10000 to 9000 BCE (or -10000 to -9000),
- 0 to 100 CE, and
- 1960 to 1961 CE.

I recommend letting R help you, but you will need to write it out on paper. Write down these calculations with `pen(cil)` and paper, and take a picture of it. As with all math and word problems,

- start with a formula, in this case, ($\#eq-r$),
- make the relevant substitutions,
- perform the calculation, and
- thus show your work.

$$\begin{array}{lcl}
 t=100 ; t+h=200 & | & r_t = \frac{\ln\left(\frac{N_{t+h}}{N_t}\right)}{h} \\
 h = (t+h) - t = 100 & | & \\
 N_t = 236904172 & | & \\
 N_{t+h} = 240611571 & | & r_{100} = \frac{\ln\left(\frac{240611571}{236904172}\right)}{100} = 0,000155
 \end{array}$$

Figure 6.5: Here I show my work to calculate r for the interval from 100 CE to 200 CE. I used my R script below to extract values from the data set and perform arithmetic.

I would use the following code to give me the data I would need for the interval for the years 100 CE to 200 CE

Specify the times and the interval you want.

```
## Get the right time points and find h
t <- 100
# recall h = (t+h) - t
t.plus.h <- 200
h <- t.plus.h - t
t; t.plus.h; h
```

[1] 100

[1] 200

[1] 100

Grab the values of N , given the years you specified.

- Find a value of `est.N` when year equals 100
- Find a value of `est.N` when year equals 200

```
## in base R
Nt <- d$est.N[d$Year == t]
Nth <- d$est.N[d$Year == t.plus.h]
Nt; Nth
```

[1] 236904172

[1] 240611571

Overthinking: You could use the `tidyverse` syntax to get the same values.

```
## using tidyverse syntax
Nt <- d %>% # use d
  filter(Year == t) %>% # use just values in this year
  select(est.N) %>% # select just N
  as.numeric() # convert from a data frame to a numeric variable

Nth <- d %>% # use d
  filter(Year == t.plus.h) %>% # use just values in this year
  select(est.N) %>% # select just N
  as.numeric() # convert from a data frame to a numeric variable
Nt; Nth
```

```
[1] 236904172
```

```
[1] 240611571
```

Do calculations to find r , the instantaneous per capita population growth rate. Note that in R `log()` refers to the natural logarithm, while `log10()` refers to the common logarithm. For more information, type `?log`.

```
# the natural log of the ratio of population sizes
logNth.over.Nt <- log(Nth / Nt)
logNth.over.Nt
```

```
[1] 0.01552817
```

Last, we find r for this particular time step of 1000 y.

```
# finding r
r <- logNth.over.Nt / h
r
```

```
[1] 0.0001552817
```

You can use R as a calculator this way to make your pen and paper work easier.

6.4.3 Programming: Use a for-loop to do all the calculations

A benefit of programming is that we can create a statement to do all our work for use. A “for-loop” is a programming trick to do a repeated process, for instance, *for* each row of a dataset.

Below is code to calculate every value of r in the entire data set, except the last year. We have to exclude the last year because we don’t know how fast the population grew from the last year to the subsequent year.

1. First we create an empty vector into which we will put our calculations of r , one at a time.
2. Next, we calculate r for each year, except the last.
3. Finally, we combine our vector of r_t with the original data.

Here we, create an empty vector of missing values. To start, this will have the same number of rows as our data set, `d`.

```
# create an empty vector of missing values.  
r <- rep(NA, nrow(d))
```

Now we calculate r **for** each row of data except the last year. Here the letter i stands in for the number of each row.

```
# Do the following steps FOR each row of our data.  
# Note we start with row 2, and use row 2 and row 1....  
for(i in 2:nrow(d)) {  
  # the second, more recent, year  
  t.plus.h <- d$Year[i]  
  # the previous year  
  t <- d$Year[i-1]  
  
  # the time interval in years  
  h <- t.plus.h - t  
  
  # the more recent N  
  Nth <- d$est.N[i]  
  # the previous N  
  Nt <- d$est.N[i-1]  
  
  # the natural log ratio  
  logNth.over.Nt <- log(Nth/Nt)
```

```
# r, which we put in the appropriate slot, year t
r[i-1] <- logNth.over.Nt/h
}

# last, let's check the result
glimpse(r)
```

```
num [1:259] 0.000237 0.000254 0.000279 0.000321 0.000368 ...
```

Finally, we combine this with our original data set. We use `cbind()` which stands for “bind columns”.

```
d.r <- cbind(d, r=r)
```

If you examine the last three rows of these data, you’ll see that we don’t have an estimate of r_t for the last year because we can’t calculate it without know the next year’s population size.

```
d.r[257:259,]
```

	Year	est.N	r
257	2019	7764951040	0.009740210
258	2020	7840952832	0.008678302
259	2021	7909295104	NA

From here on out, **we are interested mostly with the growth rate, r_t** , so I will remove the last row of the data set, or keep only the first 258 rows.

```
d.r <- d.r[1:258,]
```

6.5 Changes in human population growth rate

Let’s graph our data to examine how the *per capita* growth rate of humans changed since the agricultural revolution. I’ll pair them with population size.

```
# population size, using the first data frame
ggplot(data=d.r, aes(x=Year, y=r)) + geom_line() +
  geom_point() + labs(title="Per capita growth rate")
```

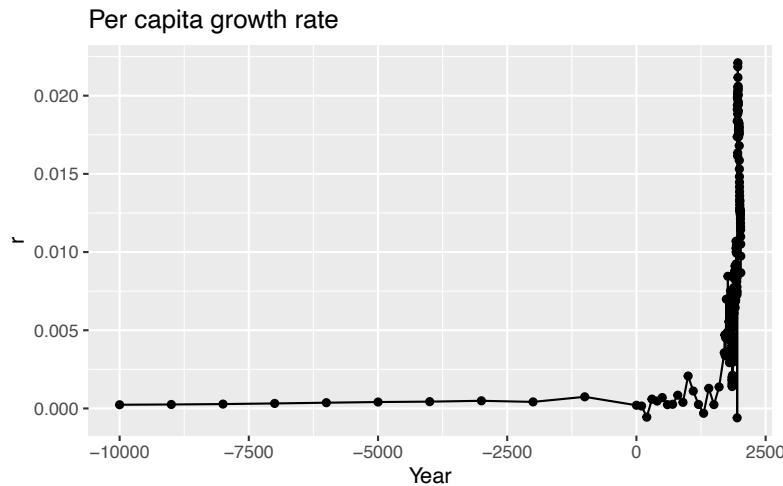


Figure 6.6: Estimates of per capita growth are positive for nearly every time interval since the agricultural revolution.

This shows us the growth rate didn't change much before the current era. However, it is hard to see what has been going on more recently. Therefore, let's plot only the data starting in year 1.

```
# filter in all years greater than or equal to 1
d.r.y1 <- d.r %>% filter(Year >= 1)

ggplot(data=d.r.y1, aes(x=Year, y=est.N)) + geom_line() +
  geom_point() + labs(title="Population Size")

ggplot(data=d.r.y1, aes(x=Year, y=r)) + geom_line() +
  geom_point() + labs(title="Per capita growth rate since 1 CE")
```

6.5.1 Your own interval

Make a graph of per capita growth rate for a time period of your own choosing. Select a time that is different than any other that we have shown here, and make it at least 10 years

6.5.2 How does per capita growth rate change with population size?

Large populations differ from small populations in a lot of ways. Large populations might make more waste products and do more damage to their own environment, and they also foster more interactions among individuals that might have positive or negative consequences.

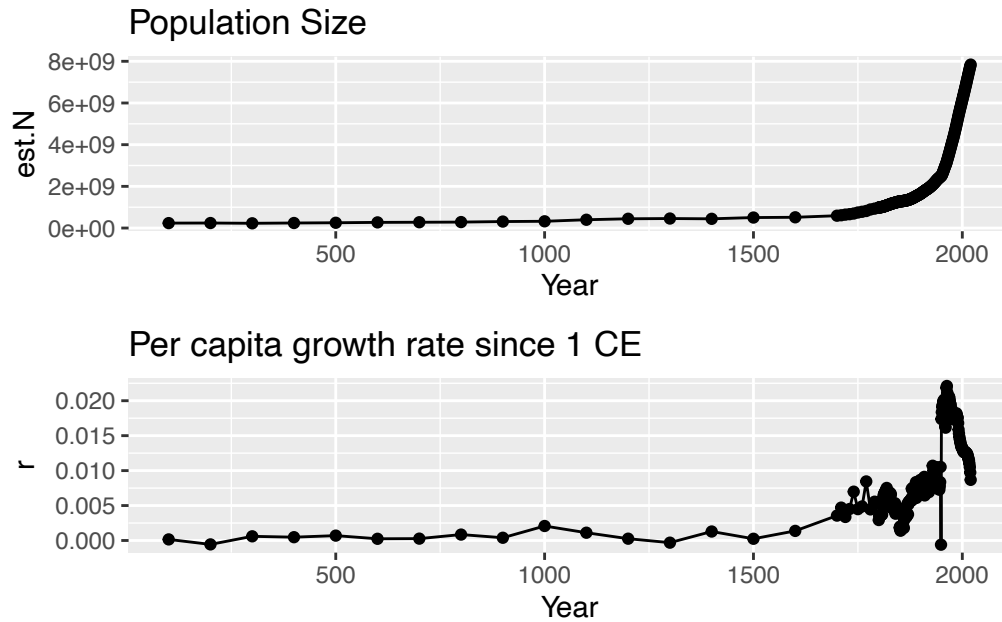


Figure 6.7: In the current era, per capita growth remains mostly positive, and then rises sharply and begin to fluctuate dramatically after 1500.

Let's make a graph of per capita growth rate as a function of population size.

6.6 Projecting into the future

Here we use our projection equation from above ($N_t = N_0 e^{rt}$) to make predictions about the future. Imagine a population of 100 individuals and $r = 0.02$ per year, and predict how big that population would become after thirty years. Assume that r is constant over that time period.

$$N_t = N_0 e^{rt} \quad (6.3)$$

$$= 100e^{0.02 \cdot 30} = 182.21 \quad (6.4)$$

When we take this approach, we are assuming that the population *continues* to grow at this rate over the projected time period. That can be a reasonable assumption over short time periods, but not over long time periods. Of course, “short” and “long” are both relative, and depend heavily of the context.

What can we learn from (Figure 6.8)? Over what time periods would it seem reasonable to assume that the human population grows at a constant rate?

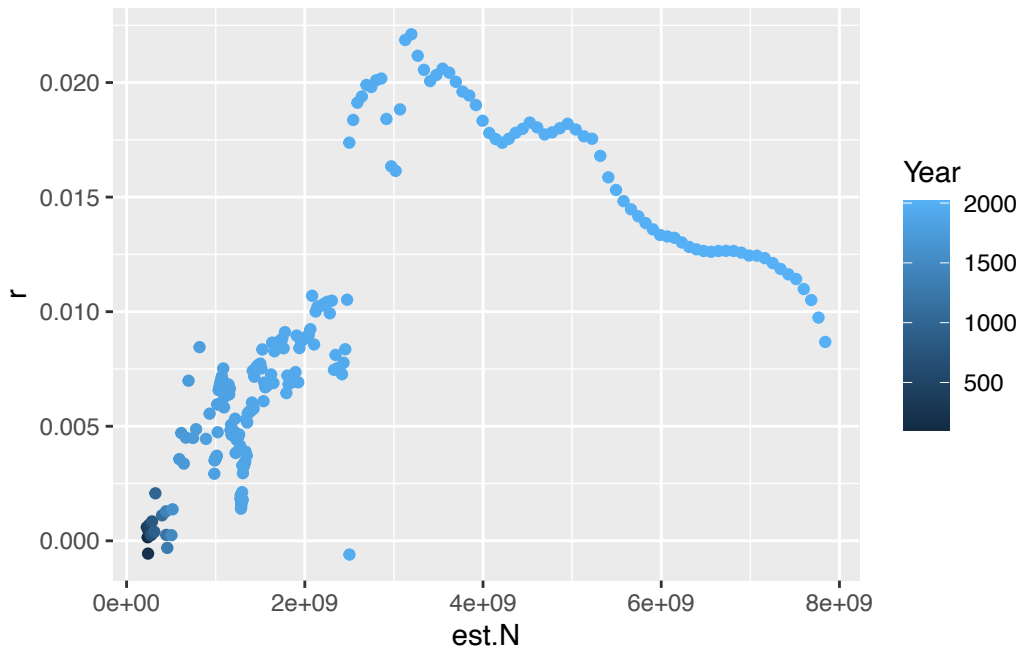


Figure 6.8: Per capita growth rate vs. population size, since 1 CE after 1500. What do you make of this? :)

6.7 Deliverables

Turn in two documents: (i) your R script, and (ii) a .docx or PDF formatted file containing the following:

1. Calculate by hand the exponential growth rate (r) for the years listed above in Section 6.4.2. Write out the formula, substitute the values to show you know where they belong, and then finish with the arithmetic to find r .
2. A graph showing per capita growth rate for a time period of your own choosing, from Section 6.5.1.
3. If the human population continued to grow at the rate you calculated for 2020, how large would the population be in 2050? Calculate this and show your work.
4. What assumptions did you make in answering Question 3? Based on results for questions 1 and 2, is it likely that the human population will reach the size that you calculated for 2050? Why or why not? Use **evidence** in the data and the graphs above to justify your explanation.
5. How many people were alive when you were born? How many people will there be at the end of your lifetime? Are you going to live through a doubling of the human population? Why or why not?

7 Structured demography: How a graduate student helped save the loggerhead sea turtle



Figure 7.1: The loggerhead sea turtle is listed as “vulnerable” by the IUCN 3.1. image: By ukanda - originally posted to Flickr as Loggerhead turtle, CC BY 2.0, <https://commons.wikimedia.org/w/index.php?curid=10481668>

7.1 Goals

In this chapter, you’ll connect sea turtle natural history to a demographic model of population growth and learn how we all learned about how to help sea turtles. Along the way, you’ll learn about sea turtle natural history, demography, dynamics and model sensitivity.

If you successfully complete this chapter, you will

- discover key stages in a sea turtles life,
- describe *population structure*,
- use natural history information to parameterize a mathematical model to make predictions about sea turtles,
- distinguish between *transient dynamics* and long-term *asymptotic dynamics*, and
- recommend management options for sea turtles conservation.

7.2 Preparatory steps

- Watch the lecture video on [Structured population growth](#), and the preceding population ecology videos (distribution and abundance, and density-independent population growth).
- Download [Crouse et al. \(1987\) from our article archive](#).
- This chapter does not use a separate data set, but it does use demographic natural history information in Crouse et al. (1987).
- Find out what you need for the deliverables (Section 7.9).

7.3 Demography

Demography includes the structure of a population which is the relative numbers of individuals of different ages or stages. Demography also includes the rates that control that structure which are primarily age- and stage-specific birth rates, death rates, and survival. We use this detailed information to make informed predictions about the population's future state, including its overall size and its structure. Therefore, demography also includes the overall rates, including population growth, that emerge from all the detailed age/stage-specific information. The study of human demography helps nations make informed policy decisions that take into account how societies will change over time due to populations that are growing, shrinking, aging or getting younger on average (Lee 2011). The study of many rare, threatened or vulnerable species includes *demography* because demography helps us understand and manage populations (Caswell 2001).

7.4 Loggerhead sea turtles

The loggerhead sea turtle (*Caretta caretta*) is a widely distributed (i.e. cosmopolitan) species that lives in oceans around the world. The International Union for Conservation of Nature (IUCN, <https://www.iucnredlist.org/>) lists it as “vulnerable” to extinction, which, sadly, is pretty good these days. The fact that the species is not worse off is in part due to research like

that of Crouse, Crowder, and Caswell (1987). This research by Deborah Crouse, Hal Caswell, and many others is the subject of what you will work with in this assignment.

Sea turtles live in warm oceans around the world and reproduce on sandy beaches. Females return to the same beaches most years to lay eggs in clutches. They come ashore, dig a hole in the sand, lay 10-100 eggs, and cover the nest, often disguising it from terrestrial predators who would like to dig them up to eat them. The female then returns to the sea to feed and avoid enemies. If everything goes right, the eggs hatch after incubation, hatchlings emerge together, struggle to the surf, and attempt to avoid the many predators (e.g., gulls, raccoons, crabs, people). Sea turtle hatchlings face many threats, including from humans who eat the eggs, and trucks or off-road vehicles can destroy the nests. In addition, lighting from nearby streets and homes can confuse hatchlings as they may crawl toward the lights as opposed to heading out to sea.

7.4.1 Questions about loggerhead survival

Include answers to these in Deliverables. 1. Consult Table 3 (Crouse et al. 1987). a. Which two stages have the lowest annual survivorship? a. Which stages have the highest annual survivorship? 1. Why do you suppose that 1st year remigrants have such low fecundity?

Fishing trawlers, which drag nets for long distances to catch shrimp and fish create another source of mortality in juveniles and adults. The sea turtles get caught in these nets and drown. These dead turtles are referred to as “by-catch”, that is, animals killed unintentionally while fishing for other species.

Sea turtle populations have been on the decline for well over a century, and some species are more endangered than others. One common conservation strategy has been to protect the eggs and hatchlings. More recently, some countries have passed laws requiring fishing trawlers (fishing boats dragging huge nets for hours at a time) to use turtle excluders devices (TEDs). These spring-loaded devices fit at the end of a trawler net and will open up when they receive a big bump from a heavy sea turtle. This allows the turtle to escape, and the fishermen lose only a fraction of their catch.

No one really knew the value of TEDs until Deborah Crouse and others modeled sea turtle populations and showed clearly that (1) saving eggs and hatchlings was not enough, and (2) increasing survival of large turtles (large juveniles and adults) was required to save the turtles from continued declines and extinction.

7.5 Demographic modeling

We use a square matrix to summarize a species’ age- or stage-specific demographic rates. A *matrix* is mathematical array of numbers with specified rows and columns. The *dimension* of a

matrix is its number of rows and number of columns. A matrix with the same number rows as columns is called a *square matrix*.

The effect of a matrix element on long term asymptotic growth rate (λ) is known as sensitivity or elasticity, depending on the details of the calculations. There are abstract ways to calculate those using linear algebra, but here we do it the easy and transparent way. We will change annual survivals of particular stages, and then recalculate λ and r to examine the effect directly.

Crouse, Crowder, and Caswell (1987) performed a demographic analysis of an endangered sea turtle species, the loggerhead (*Caretta caretta*). Management of loggerhead populations seemed essential for their long term survival, and a popular management strategy had been and still is to protect nesting females, eggs, and hatchlings. The ground breaking work by Crouse¹ and her colleagues compiled data to create a stage-based projection matrix to analyze quantitatively which stages are most important and least important in influencing long-term growth rate. This work led to US Federal laws requiring that US shrimp fishermen use nets that include Turtle Excluder Devices (TEDs).

A *demographic transition matrix* is merely a compact quantitative summary of a species' life history events. It includes the annual probabilities that an individual remains in a particular stage, grows into a later stage, or even regresses back to an “earlier” stage. It also includes the fertility or number of progeny it contributes to the first stage. These stages may be age classes (e.g., 0-5 y), sizes (e.g., 2-4 leaves), or some other major life history stage (e.g., first year breeder).

The transition matrix holds each of the probabilities and fertilities in a way that the column identifies the stage at time t whereas the row identifies the next time period $t + 1$.

In Figure 7.2, the element s_2 is the probability that a juvenile in year 1 survives to remain a juvenile in year 2. The element g_2 represents the probability that a juvenile in year 2 survives and also become a reproductively mature adult in year 2. The element g_1 is the probability that a hatchling survives and becomes a juvenile in year 2. The element s_3 is the probability that an adult survives and remains an adult. We refer to the elements on the diagonal (s) as **stasis** and the elements on the off-diagonal (g) as **growth**.

Fecundity or reproduction is seen as adults contributing the to smallest or youngest stage. The element f_3 is the number of hatchlings that occur in year 2 based on the survival and reproduction of the average adult. The element s_1 is the survival of hatchlings. If the stage only exists for one year, then $s_1 = 0$ because, by definition, after one year a hatching is a juvenile, or it is dead.

¹Crouse was a graduate student at the time—graduate students are the life-blood of modern science, doing cutting edge work and pushing their fields forward.

		Year 1		
		H_1	J_1	A_1
Year 2	H_2	s_1		f_3
	J_2	g_1	s_2	
	A_2		g_2	s_3

Figure 7.2: A toy demographic transition matrix, with stages of hatchlings (H), juveniles (J), and adults (A). Elements represent the probability that an individual belonging to the stage in the column stays in or transitions to the stage in the row over a single year. Fecundity (f_3) represents the total number of offspring that an individual contributes to the stage in that row.

7.5.1 Connecting life history information to a model

Crouse, Crowder, and Caswell (1987) gathered data from a variety of sources to estimate loggerhead turtle survival and fertility at different ages and stages. They organized these data in the form of a transition matrix, **A**. They then used this model to evaluate the relative importance of different stage transitions to the population growth rate.

On the basis of their analysis, they were able to identify that the survival of juveniles and subadults was the most important determinant of turtle growth rate, under the current circumstances. It was this insight, made possible only because of a structure population model, that led to changes in laws to require the use of turtle excluder devices (TEDs) in fishing and shrimp trawlers.

In this assignment, you will walk through the types of analyses done by Deborah Crouse and her colleagues. They combed the literature to find information about survival and reproduction by loggerheads and summarized those findings in their tables 1 and 2. They then organized that information into a more condensed form and presented it in Table 3, shown here in Figure 7.3.

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TABLE 3. Stage-based life table for loggerhead sea turtles based on data in Frazer (1983a). These values assume a population declining at $\approx 3\%/yr$.

Stage number	Class	Size* (cm)	Approximate ages (yr)	Annual survivorship	Fecundity (no. eggs/yr)
1	eggs, hatchlings	<10	<1	0.6747	0
2	small juveniles	10.1–58.0	1–7	0.7857	0
3	large juveniles	58.1–80.0	8–15	0.6758	0
4	subadults	80.1–87.0	16–21	0.7425	0
5	novice breeders	>87.0	22	0.8091	127
6	1st-yr remigrants	>87.0	23	0.8091	4
7	mature breeders	>87.0	24–54	0.8091	80

* Straight carapace length.

Figure 7.3: Estimates of stage-specific survival and fecundity for loggerhead sea turtles, from Table 3 in Crouse et al. (1987). Based on information described in Tables 1 and 2 in Crouse et al. (1987).

Their transition matrix (Table 4):

$$\begin{matrix}
 & \begin{pmatrix} H & J_s & J_l & sub & B_1 & B_2 & M \end{pmatrix} \\
 \begin{pmatrix} H \\ J_s \\ J_l \\ sub \\ B_1 \\ B_2 \\ M \end{pmatrix} & \begin{pmatrix} 0 & 0 & 0 & 0 & 127 & 4 & 80 \\ 0.6747 & 0.7370 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.0486 & 0.6610 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.0147 & 0.6907 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.0518 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.8091 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.8091 & 0.8089 \end{pmatrix}
 \end{matrix}$$

7.5.2 Connecting life histories to demographic matrices

1. Compare Table 3 and Table 4 (Crouse et al. 1987). What are the relationships between survival and fecundity in Table 3, and the diagonal (stasis) and off-diagonal elements (growth) of the matrix in Table 4? How are they related?
2. Why are the diagonal matrix elements (Table 4) not the same as the annual survivorship in Table 3?

To use the information in Table 3 to make predictions about the long-term viability of the population and to make recommendations for the conservation of the population, Crouse and colleagues need to combine these stage-specific demographic rates into a model, and summarize the behavior of that model.

Their first step was to put the above rates into a demographic transition matrix, where columns represent stages in one year, and the rows represent stages in the next year.

7.6 Analysis of a demographic matrix

As with all of our work, confirm you are in the correct working directory.

```
getwd()
```

```
[1] "/Users/stevenmh/projects/intro_to_ecology"
```

Fix this, if necessary.

Load needed packages.

```
## for graphics and data manipulation
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v ggplot2    4.0.2      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Two of the techniques we use to analyze a demographic matrix are **simulation**, and a mathematical technique called **eigenanalysis**. Simulation allows us to reveal the dynamics through time, year after year, generation after generation. Eigenanalysis allows us to very quickly quantify average characteristics of the population, including the population growth rate, λ , the stable stage distribution, and the elasticities.

- **population growth rate**, λ , is the long term *finite rate of increase* and assumes density-independent growth.
- **stable stage distribution** is the set of expected relative densities of the different stages.
- **elasticities** are the relative importances of the different transitions to determining the finite rate of increase, λ .

We will focus on lambda, and do some experiments playing around with survival and growth.

First, we need to create the observed transition matrix.

```
A <- matrix(
  c(0,      0,      0,      0,      127,      4,      80,
    0.6747, 0.7370, 0,      0,      0,      0,      0,
    0,      0.0486, 0.6610, 0,      0,      0,      0,
    0,      0,      0.0147, 0.6907, 0,      0,      0,
    0,      0,      0,      0.0518, 0,      0,      0,
    0,      0,      0,      0,      0.8091, 0,      0,
    0,      0,      0,      0,      0,      0.8091, 0.8089),
  nrow=7, ncol=7, byrow=TRUE)
rownames(A) <- colnames(A) <- c("H","J_s","J_l", "sub", "B_1", "B_2", "M")
A
```

	H	J_s	J_l	sub	B_1	B_2	M
H	0.0000	0.0000	0.0000	0.0000	127.0000	4.0000	80.0000
J_s	0.6747	0.7370	0.0000	0.0000	0.0000	0.0000	0.0000
J_l	0.0000	0.0486	0.6610	0.0000	0.0000	0.0000	0.0000
sub	0.0000	0.0000	0.0147	0.6907	0.0000	0.0000	0.0000
B_1	0.0000	0.0000	0.0000	0.0518	0.0000	0.0000	0.0000
B_2	0.0000	0.0000	0.0000	0.0000	0.8091	0.0000	0.0000
M	0.0000	0.0000	0.0000	0.0000	0.0000	0.8091	0.8089

Then we can calculate lambda. Remember that lambda is just the ratio of population sizes in sequential years ($\lambda = N_{t+1}/N_t$).

```
original_lambda <- Re( eigen(A)$values[1] )
original_lambda
```

[1] 0.945031

This value, $\lambda = 0.94$, is less than $\lambda = 1$, and therefore indicates a shrinking population. **This shows that the loggerhead sea turtle was predicted to decline.**

What can we do?

7.7 Finding the life history stage that could make the biggest difference

Part of managing rare species is figuring out which part of the population, the age, or **the stage on which to concentrate your efforts.**

A good way to figure out where to focus our effort is to find out which stage could make the biggest difference to the growth rate, λ . If we improve survival of an stage and it helps λ increase above 1, focusing on that stage could be a good strategy.

Your strategy will be to improve the survival in each stage so that individuals in that stage do not die. They either survive and remain in that stage, or they grow into the next stage. That means that with perfect survival, stasis and growth sum to nearly one.

For each stage, we will pretend that we can improve annual survival 99.9%. Some of our stages are only one year in duration, while others are 7, 8, or 30 years in duration. This makes it a little tricky to calculate our new simulated matrix elements. I will do that part for you.

In each of seven simulations, you will test the effect on growth rate (λ) of improved survival.

7.7.1 Helping hatchlings

For your first experiment, you will test the effect on lambda of improving survival and growth of eggs and hatchlings to 99.9%, or annual survivorship of 0.999. Here are the steps:

1. Copy the old matrix and rename it (see below!).
2. Change the new matrix to include survivorship of 0.999 for the hatchlings (see below!).
3. Calculate lambda for the new matrix (see below).

```
## new matrix
A1 <- A

## Change just the hatchling survival/growth
A1[2,1] <- 0.999
```

```
## look at the new matrix
A1
```

	H	J_s	J_l	sub	B_1	B_2	M
H	0.000	0.0000	0.0000	0.0000	127.0000	4.0000	80.0000
J_s	0.999	0.7370	0.0000	0.0000	0.0000	0.0000	0.0000
J_l	0.000	0.0486	0.6610	0.0000	0.0000	0.0000	0.0000
sub	0.000	0.0000	0.0147	0.6907	0.0000	0.0000	0.0000
B_1	0.000	0.0000	0.0000	0.0518	0.0000	0.0000	0.0000
B_2	0.000	0.0000	0.0000	0.0000	0.8091	0.0000	0.0000
M	0.000	0.0000	0.0000	0.0000	0.0000	0.8091	0.8089

In our new matrix, we see that 99.9% of hatchlings (first column, H) survive to grow into being small juveniles that next year (second row, J_s), that is, $a_{2,1} = 0.999$.

Now calculate lambda of the new matrix.

```
## using Re() extracts just the real part of the eigenvalue, and not the imaginary part
Re( eigen(A1)$values[1] )
```

```
[1] 0.964903
```

This is the new growth rate. Is it greater than 1? Does it suggest that improving hatchling survival will allow this turtle population to grow? How does this compare to the original value of lambda of 0.94? That is, *how much did saving all the hatchlings help the loggerhead population?* You'll need this for deliverables.

7.7.2 Helping small juveniles

Here you will test the effect on lambda of improving survival and growth of small juveniles to 99.9%, or annual survivorship of 0.999. This stage lasts seven years, and so I calculated the new matrix elements for you (see the code).

Here are the steps:

```
## new matrix with NEW NAME, A2
## make sure you start with the ORIGINAL matrix A
A2 <- A

## Change just the stasis and growth for small juveniles
A2[2,2] <- 0.857 # stasis
```

```
A2[3,2] <- 0.142 # growth
A2
```

```

      H    J_s    J_l    sub    B_1    B_2    M
H    0.0000 0.000 0.0000 0.0000 127.0000 4.0000 80.0000
J_s  0.6747 0.857 0.0000 0.0000  0.0000 0.0000  0.0000
J_l  0.0000 0.142 0.6610 0.0000  0.0000 0.0000  0.0000
sub  0.0000 0.000 0.0147 0.6907  0.0000 0.0000  0.0000
B_1  0.0000 0.000 0.0000 0.0518  0.0000 0.0000  0.0000
B_2  0.0000 0.000 0.0000 0.0000  0.8091 0.0000  0.0000
M    0.0000 0.000 0.0000 0.0000  0.0000 0.8091  0.8089

```

Now that you have a new matrix, calculate the lambda for this matrix. Do that now by borrowing and editing code from above.

Now calculate the finite rate of increase (λ)

```
Re( eigen(A2)$values[1] )
```

```
[1] 1.03806
```

What does this value of lambda tell you about the effect of increasing survival of small juveniles (if we could do that)?

7.8 Simulating helping other stages

Repeat a similar procedure for each stage, using these values:

Table 7.1: New values to use in the loggerhead sea turtle transition matrix, changing only one stage at a time. These changes reflect 99.9% annual survival. Stasis values belong on the diagonal, such as `A[5,5] <- 0`, and growth values belong below the diagonal, such as `A[6,5]<-0.999`. Start with the original matrix, make a copy, and then change only one stage at a time.

Stage	Stasis	Growth
H	0.000	0.999
J_s	0.857	0.142
J_l	0.875	0.124
sub	0.833	0.166

Table 7.1: New values to use in the loggerhead sea turtle transition matrix, changing only one stage at a time. These changes reflect 99.9% annual survival. Stasis values belong on the diagonal, such as $A[5,5] \leftarrow 0$, and growth values belong below the diagonal, such as $A[6,5] \leftarrow 0.999$. Start with the original matrix, make a copy, and then change only one stage at a time.

Stage	Stasis	Growth
B_1	0.000	0.999
B_2	0.000	0.999
M	0.966	0.000

Record the new value of lambda for each separate simulation.

7.9 Deliverables

Put the following into a single document:

1. A life cycle graph for the loggerhead population. On your figure, include the numerical elements associated with each transition, based on the original matrix.
2. A table similar to Table 7.1 with all the stages, the provided values of stasis, and growth, and your new values of lambda.
3. Assume it is your interest to maximize long-term population growth rate (λ), for which stage or stages should you invest in protection measures? Explain why, and use your values of lambda as evidence in your answer.

Upload this document and your R script to the appropriate site.

8 Density-dependence in population growth

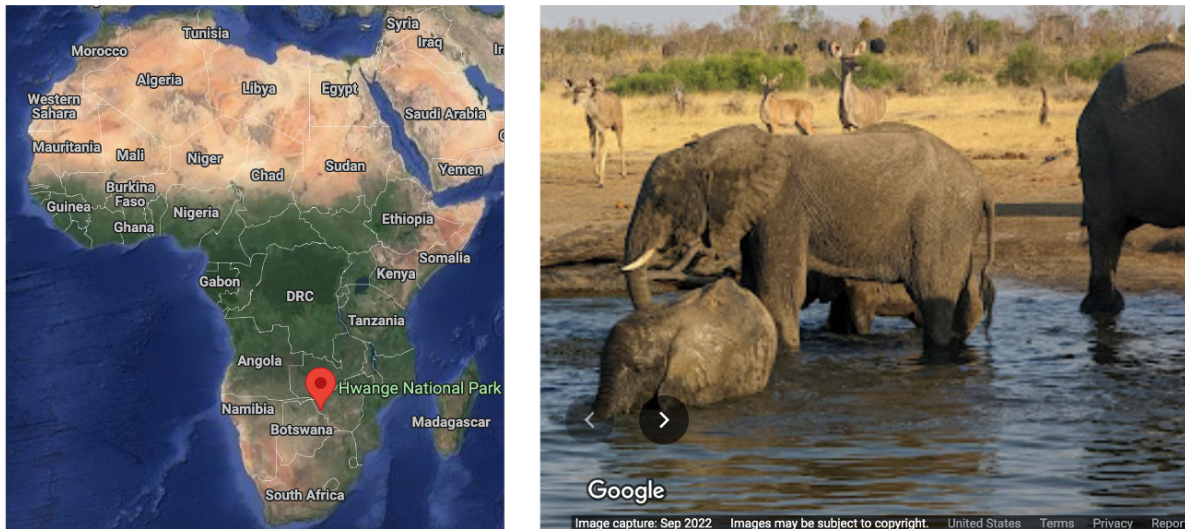


Figure 8.1: Elephant density in Hwange National Park, Zimbabwe, depends strongly on annual rainfall (Chamaillé-Jammes et al. 2008).

8.1 Goals

In this chapter, you'll learn how we represent crowding in populations. Per capita population growth rate often depends on population density. Along the way, you'll learn we take ideas seriously, and follow them to their logical conclusions.

If you successfully complete this chapter, you will

- extend exponential growth to include the effect of crowding,
- define *density-dependence*,
- distinguish between *transient dynamics* and long-term *asymptotic dynamics*, and
- discover the consequences of crowding by *solving* a rate equation.

8.2 Preparatory steps

- Watch the lecture videos on [density-dependent population growth \(parts 1 & 2\)](#). These videos assume that you watched the preceding population ecology videos.
- Find out what you need for the deliverables.

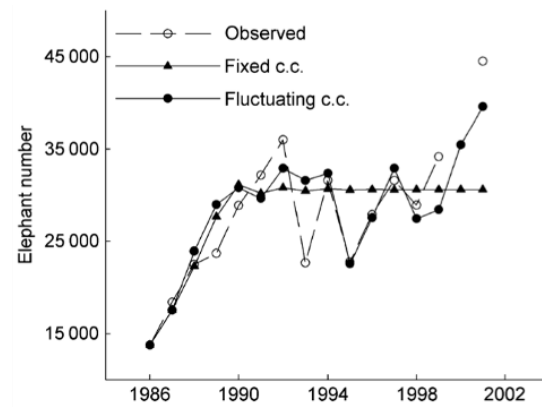
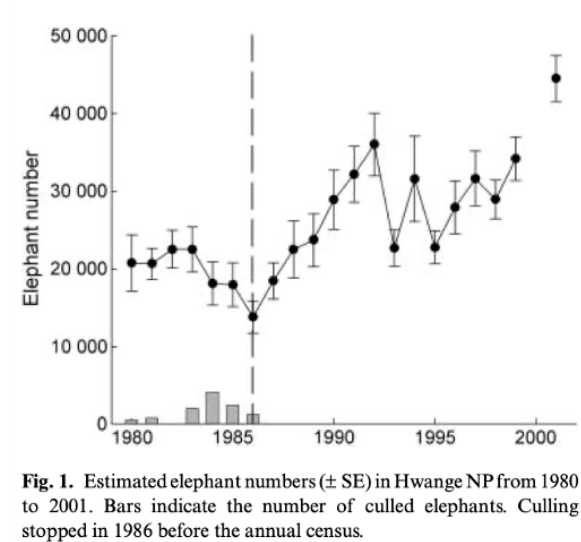


Figure 8.2: The Hwange elephant population grew dramatically after the end of hunting (“culling”) that had kept the population size small (Chamaillé-Jammes et al. 2008).

8.3 Models as assumptions

Continuous logistic population growth is just a set of assumptions about how a population grows. The rate of change in our population is determined by its birth rate and death rate, and how population size affects these.

The symbol dN/dt means “change in population size (N) per unit change in time (t)”.

Recall exponential growth (J-shaped growth) is

$$\frac{dN}{dt} = rN$$

Logistic growth adjusts exponential growth by penalizing it for population size.

$$\frac{dN}{dt} = rN(1 - \alpha N)$$

where α is a small number much less than one, and it is the effect of one individual on the growth rate of the population. Thus, as N gets bigger, $\alpha N \rightarrow 1$ (approaches 1), and $1 - 1 = 0$.

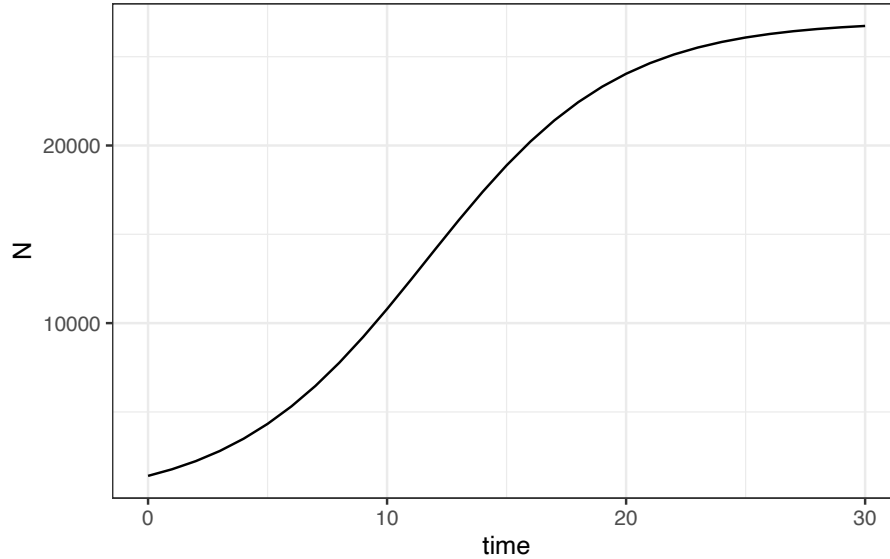


Figure 8.3: Logistic growth in a hypothetical population (parameters: $r = 0.25$, $\alpha = 1/27000$).

You see that in Figure 8.3, the growth rate slows dramatically as the population size approaches some limit. We call that limit the **carrying capacity** of the population, and it is determined by the effects of individuals on the population growth rate, or α .

8.3.1 Math describes nature: on parameters and their meanings

Often, we can describe a biological process in several different mathematical ways. Density-dependence in logistic growth is one such case.

We can define logistic growth in two ways that are opposite sides of the same coin. Above, we used α to quantify the average effect of one individual on the growth rate of the population. Another common way to define logistic growth is by the consequences of these average individual effects: the carrying capacity which we call K . It turns out that the carrying capacity is the inverse of the effect of an individual, so that

$$K = \frac{1}{\alpha}$$

and as a result we can rewrite logistic growth as

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right)$$

which is exactly equivalent to the first version we described above.

8.3.2 Solving a rate equation

We **solve** a rate equation in order to learn something about its **asymptotic** behavior. By “learn something” we mean figure out what will determine the size of population. By “asymptotic”, we mean really, really, long-term. We hope to find out what will eventually happen to the population and what will control its destiny.

Another way to think about this is that we are seeking to find the **equilibrium state**, the state of the population toward which the population will tend to move. Imagine a marble in a wok or bowl. The marble will tend to move toward the bottom. When it finally settles down and *stops moving*, it is at its equilibrium state.

We solve a rate equation by setting it equal to zero, because **we want to understand what determines when the population stops changing**.

8.3.3 Assignment 1: Use math to understand the consequences of our assumptions.

Now solve for the equilibrium, when $dN/dt = 0$. That is, let

$$0 = rN(1 - \alpha N)$$

and solve for N . That is, rearrange to get N on the left side $N =$ and other stuff on the right side, and then simplify. **SHOW ALL THE STEPS IN YOUR WORK.**

8.4 Discrete difference equations

Sometimes, it is simpler to think about rate equations in a step-wise annual fashion, where each change is a **discrete** step, rather than continuous change.

N (this year) = N (last year) + births - deaths

In this case, the rate of change is a **difference** between N this year versus last year,

N (this year) - N (last year) = ΔN = births - deaths

In this situation, we call this a **difference equation**.

The difference equation for logistic growth is

$$N_t - N_{t-1} = \Delta N = r_d N_{t-1} (1 - \alpha N_{t-1})$$

where the number of births and deaths is described by $r_d N_{t-1} (1 - N_{t-1}/K)$.

Notice an important feature of this equation: the density-dependence $(1 - \alpha N_{t-1})$ depends on the size of the population at one point in time $(t - 1)$, but the population is projected out past this time. As a result, the dampening or slowly effect of density is *delayed*, always lagging behind just a little bit. You will investigate the consequences of this in your exercise.

We can project the population dynamics using

$$N_t = N_{t-1} + r_d N_{t-1} (1 - \alpha N_{t-1})$$

8.4.1 Assignment 2: Explore delayed density-dependence

Here you will explore the consequences delayed density-dependence. In this exercise, you will discover *stable* points in a **stable limit cycle**. These *stable points* are the values of N to which a population returns.

You will also discover mathematical **chaos**. In science, **chaos** is defined as a deterministic, non-repeating pattern that is bounded, and characterized by sensitivity to initial conditions.

In the code below, you will do several things. First, we set up things that won't change,

1. the carrying capacity ($K = 100$),
2. the number of generations (50).
3. an empty vector that will eventually hold all of our values of N .
4. the initial size of the population. For simplicity, keep this between 40 and 70.

Next we do things to investigate the consequences of delayed density-dependence. Here we investigate what happens when populations grow slowly (small r_d) vs. quickly (large r_d).

You will do these things four times.

1. Select a value of r_d . Pick four different values for r_d , one each between
 - a. 0.2 and 1.5,
 - b. 2.1 and 2.4,
 - c. 2.45 - 2.56
 - d. 2.7- 3.
2. For your first value of r_d , project the dynamics, graph the time series of the dynamics, and save the graph under a unique name.
3. Repeat for the second, third, and fourth values of r_d .

When you are finished, you should have four graphs that each use a different value of r_d and each should look quite different.

Step 1 Set up things that won't change.

```
## Define K, NO, the number of generations
K <- 100
gens <- 50
## make numeric vector for each N in each generation, plus one more
N <- numeric(gens)

## Start the population out at this number of individuals
N[1] <- 45
```

Select a value of r_d and use a for-loop to project the population through time, from one year to the next.

```
#####
## THIS IS WHERE YOU PICK THE VALUE OF rd YOU WANT
rd <- 0.2
#####

## project the population
for(t in 2:gens) {
  N[t] <- N[t-1] + N[t-1]*rd*(1-N[t-1]/K)
}

# store this result in a data frame
my.N <- data.frame(year=1:gens, N=N)

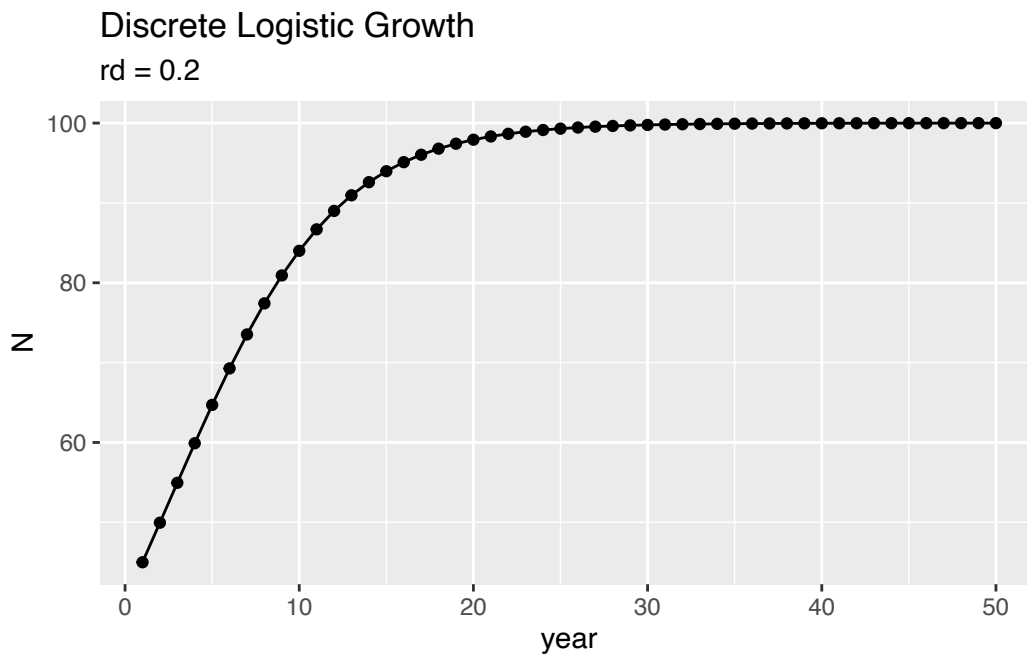
#for kicks, look at rows 1-3.
my.N[1:3,]
```

	year	N
1	1	45.00
2	2	49.95
3	3	54.95

Graph the time series of the population dynamics, and **change the name of the file** that you save using `ggsave()`.

```
## graph your data
ggplot(data=my.N, aes(x=year, y=N )) +
  geom_point() +
  geom_line() +
```

```
labs(title = "Discrete Logistic Growth",
      subtitle=paste("rd =", rd))
```



```
## save your graph USING A UNIQUE NAME
ggsave("dynamics-rd-0.2.png", height=5, width=6)
```

Put into one document your four graphs, and a description of how the graphs differ, including

- the relationship between r_d and the year in which the carrying capacity is reached or exceeded,
- the predictability of the dynamics, and
- the number of stable points to which the population returns.

8.5 Deliverables

Put your work from section 8.3 (Assignment 1) and section 8.4 (Assignment 2) into one document and turn in by the due date.

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